Nodes and Terminals Facility-Class Design Basis

The First Facility Class of the Transportation Sector Design Basis

DBA-TRAN-ND-FC1 — Facility-Class Design Basis, DBA-TRAN Series

Issued Under: Transportation Sector Framework (DBA-TRAN_Sector_Framework v0.11); DBA-MA Parent Framework

Version 0.3 — DRAFT

October 2026

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Working Draft — Released for Review and Discussion

SUGGESTED CITATION

T. Roxey, "DBA-TRAN-ND-FC1 — Nodes and Terminals," Version 0.3, in CIKR Design Basis Examples, Edition 1 (November 2026), Eclectic Technologies, November 2026. https://doi.org/10.5281/zenodo.23863986

This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

1.1 Purpose

This document sets out the design basis for the nodes of the transportation system: the airports, seaports and marine terminals, rail classification yards and intermodal terminals, transit stations and major truck terminals where movement gathers, stops and changes hands. It is the first facility class in the DBA-TRAN series. The *Transportation Sector Framework* (DBA-TRAN_Sector_Framework) states the sector's design basis in general terms, and this document applies it to the nodes. It names the events a node must withstand, traces how each reaches the people and the goods that pass through it, sets the measures that size its protection, and states who must authorize what. It describes no particular airport, port, yard or station. A design basis for a specific node starts from this document and adds only what is particular to it.

The framework names three function classes and a coordination class, and this one comes first because it is where the sector's flow is most concentrated. A small number of nodes carry a large share of the nation's movement, and the loss of one of them is not a local event. The framework records the major port and the air traffic control class as the leading candidates for a first instrument (framework §9). On the author's decision of 8 October 2026 the classes are written in the order the register numbers them, so the node comes first and the control class last, once the classes it controls have said what they need from it.

1.2 Scope

The class covers everything inside a node's boundary. At an airport that is the runways, taxiways and aprons, the passenger and cargo terminals, the gates, the baggage and cargo handling systems, the fuel distribution on the airfield, and the airport's own power, water and communications. At a port it is the berths and wharves, the ship-to-shore cranes, the container yard and its handling equipment, the gates, the warehouses and transit sheds, the on-dock rail, and the terminal operating systems that plan and track all of it. At a rail node it is the classification and hump yards, the intermodal ramps, the locomotive and car shops, and the yard's own systems. At a transit station it is the platforms, concourses, vertical circulation, fare and access control, ventilation and the station's fire and life-safety systems. At a truck terminal it is the docks, the yard, the gates and the systems that run them. The class covers the systems that monitor and manage all of these.

On the author's decision of 8 October 2026, the node stops at its fence. What lies inside the facility boundary is this class's. What lies outside it is not, even where it serves only the node, and five boundaries follow. The airspace above and around an airport, the harbor channel and its approaches, the track and road beyond the yard throat and the gate, and the tunnels that connect a station to the rest of its line belong to DBA-TRAN-WY-FC2 (*Ways and Linehaul*). The control tower, the vessel traffic service, rail dispatch and the transit control center belong to the coordination class, DBA-TRAN-CntlCo (*Traffic Management, Signaling and Control*), even

where they sit on the node, because their consequence is the authority to move and not the node's throughput. The aircraft at the gate, the vessel at the berth and the train in the yard belong to DBA-TRAN-CV-FC3 (*Conveyances and Fleet*). Petroleum, gas and chemical terminals at a port, and hazardous cargo held at any node, are shared under the release-and-network rule of Section 8.3. And where a transit station lies beneath a public-assembly venue, the station is this class's and the venue is the Commercial Facilities series' (Section 8.3). On the author's decision of 10 October 2026, the queue a node causes beyond its fence is this class's as a consequence it measures (Section 4.6), while the anchorages and holding areas where it waits are the ways class's as places and the coordination class's to manage; and the people aboard a train standing at a station's platform count within the station's occupancy, the train being the conveyance class's (Section 3.3).

1.3 One Class, Five Modes

On the author's decision of 8 October 2026, the nodes are one class, sub-indexed by mode as the framework sets out (framework §3.1, §3.6): aviation; freight rail; passenger rail and transit; maritime and inland waterways; and highway and motor carrier. A requirement in this document applies to every mode unless it names one. Where the modes differ in a way that changes what must be done, the requirement is stated for each, and nowhere else. A mode-specific instrument for the highest-consequence nodes — the hub airport and the major container port — may follow as a derivative (Section 9.2); it would add what is particular to its mode and would not restate what is common.

The modes differ most in who regulates the node and in the physics of what moves through it, and least in why the node matters. An airport, a port and a classification yard are the same functional object: a place where flow converges, is sorted and is sent on, and whose loss strands whatever cannot be sent through somewhere else.

1.4 The Reference Configurations

Four configurations are used as reference cases.

A major container port is the governing freight configuration. Its berths, cranes and yard move a large share of a region's or a nation's imports, through a terminal operating system that plans every move, and on-dock and near-dock rail and a truck gate connect it inland. The twin ports of Los Angeles and Long Beach show the type: in 2023 they handled about 10.8 million of the 40.6 million loaded twenty-foot-equivalent units that passed through 109 United States ports, a little over a quarter of the national total, and the top five ports handled half (Bureau of Transportation Statistics, *Port Performance Freight Statistics*, 2026 Annual Report; the combined share is computed from its table). It sets the class's first measure.

A hub airport is the configuration that carries both of the class's consequence chains at once. It moves tens of millions of passengers a year and a large volume of cargo, through terminals that hold thousands of people at any moment. The thirty or so large-hub airports enplane about seven

in ten of the country's airline passengers (Federal Aviation Administration, *Terminal Area Forecast Summary FY2025–2055*; the share is computed from its figures), and Atlanta alone enplaned some 52.5 million in 2024 (FAA, CY2024 enplanements).

A rail hub is the third. Chicago is the type. The Association of American Railroads has estimated that a quarter of the nation's rail traffic touches the Chicago region (as presented in the Federal Highway Administration's *Talking Freight* webinar of October 2016), and the CREATE program reports that nearly half of the nation's intermodal units pass through it (CREATE, *Overview*, April 2021). The classification yards that sort the freight have grown fewer as the railroads have consolidated; trade reporting counts thirty-six hump yards left in North America at the end of 2024, from more than 150 in 1975 (*Trains*, December 2024). The fewer there are, the more each one carries.

An urban transit station beneath a venue is the fourth, and it governs the occupancy chain. Pennsylvania Station beneath Madison Square Garden is the type the Commercial Facilities series names (*Commercial Facilities Sector Framework* §1.5). Its platforms, concourses and stairs must clear a crowd of commuters and an arriving event crowd at once, through exits that serve both.

A requirement that names one configuration applies to that configuration. A requirement that names none applies to all of them.

1.5 Class Type

This is a physical class under the author's determination of 30 September 2026 (Facility-Class Types Are Physical, Coordination and Logical). The harm it can do comes from the loss of places and things and of the people in them: a berth, a crane, a runway, a station concourse. A node is also run by systems that plan and record every move it makes, and the loss of those systems stops it as surely as a fire. Those systems are part of the node and are designed here; their authority to move anything beyond the fence is the coordination class's.

1.6 Relationship to Governing Documents

This instrument is issued under the Transportation Sector Framework and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, principles or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where this instrument develops class-level content ahead of the framework, it records in Section 10 what the framework should take up. References of the form "framework §N" are to the Transportation Sector Framework in force.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition*

and Determinism). Local sovereignty, the property of a protective function that executes independently of reachable digital authority, is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the authorization obligation may be discharged, including in advance, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*The Envelope*) reserves the term "envelope" for the pre-authorized envelope within which advance discharge is valid; this instrument uses the word in that sense only, and calls a region of normal operation whose limits are monitored an operating band. Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it. The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

A second Line applies to some actions of this class. *The Surveillance Bright Line* (FD-SBL) is the sibling of FD-BL for actions that act on persons' cognitive state, political behavior or liberty, and it is owned by Project 1. Passenger screening at a node is such an action. FD-SBL requires that a document sitting in both corpora name which definition governs the action under analysis (*The Surveillance Bright Line* §2.1), and Section 6.4 does so. FD-SBL is cited without a version token, and FD-SBL-REG (*Surveillance Bright Line Register*) is authoritative for its version state.

The communications design basis is cited, not inherited. DBA-MA-CDB (*Military Action Communications Design Basis*) sets out seven requirements, CDB-1 through CDB-7, for every facility in every sector whose communications may be degraded or denied, and Section 7 states how the seven apply to the nodes, as *Military Action Communications Design Basis* §3 requires. The design basis for timing receivers is cited from *Positioning, Navigation and Timing* §7, which holds it for every sector.

2. The Nodes and Terminals Facility Class

2.1 Defining Characteristics

Four features define the class.

First, it concentrates flow. The sector's other physical classes are spread out along routes and fleets; this one gathers what they carry into a few places. A port complex that handles a quarter of a nation's loaded container traffic, a rail hub that a quarter of its rail traffic touches and a hub airport that handles a large share of its passengers are each a single point through which much of the system passes. Their loss is a system event, and its size is set less by what the node holds than by what cannot be sent anywhere else.

Second, it is where custody and mode change. Goods arrive by one mode and leave by another, under the care of one company and then the next, recorded at each step by systems that several parties share. A node runs on information as much as on cranes and runways: what is in each container, where it is in the yard, which flight a bag belongs to, which car goes on which train.

When that information is lost, the physical node can be intact and still unable to move anything, because no one knows what to move where.

Third, many nodes share the same systems. Terminal operating systems, airport check-in and boarding platforms, and the endpoint software on their computers are sold by a few suppliers to many nodes at once, and a terminal operator may run the same system at dozens of terminals worldwide. A single failure or compromise in such a system reaches every node that runs it, at the same moment. The shared platform includes the operator's own enterprise network and identity domain wherever the node's operating systems depend on them: Maersk's terminals stopped in 2017 because they shared the company's network with an office in Ukraine (*NotPetya Attack* F-2). This is the class's common-cause failure. Two of its six reference events turn on it, and a third shared-platform event, the attack on an airline check-in system in September 2025, is cited (Section 2.3).

Fourth, many of its nodes hold crowds. A passenger terminal, a transit station or a ferry terminal holds thousands of people at once, most of them strangers to the building, many with luggage, some with children or disabilities, and all of them dependent on the node's systems to tell them where to go. The node's second consequence is life safety, and it is borne by people who did not choose to be there in an emergency and do not know the way out.

2.2 Deployment Typologies and Size Range

The class spans a wide range. At one end are the largest container ports, the hub airports and the rail hubs, each handling a national share of its mode and each operated by several companies under a public authority. At the other are a regional airport with a few flights a day, a small port with a single terminal, a short-line railroad's yard and a suburban rail station. Between them are the medium hubs, the regional ports and the intermodal terminals that serve a metropolitan area. Ownership is equally varied: an airport or port is usually owned by a public authority and operated in large part by tenants — airlines, terminal operators, stevedores, ground handlers — while a rail yard is owned and run by its railroad and a truck terminal by its carrier. Air cargo has its own hubs: Memphis and Louisville, the hubs of the two largest express carriers, were among the five busiest cargo airports in the world in 2024 (Airports Council International, preliminary 2024 rankings). The class design basis applies to all of them, and the values of its measures differ widely across them.

2.3 Consequence Context

The class's governing harm, as the framework states it, is throughput loss at a concentration point (framework §3.2). On the author's decision of 8 October 2026, its second harm is the loss of life or injury among the people a node holds. Its third is what both do to the sectors that depend on the node's flow.

Six events are the reference events for the class. Seven were proposed here and conferred by the framework at its §9.1 as DBA-TRAN-RE-1 to DBA-TRAN-RE-7; on the author's triage ruling

of 9 October 2026 the seventh, the attack on the Collins Aerospace check-in system, is withdrawn as a reference event and is cited below as an illustration. The six instruments are written, and under the determination of 9 October 2026 that gives reference events their own project they are held by Project 11. This revision takes in what they return, and the accounts below are conformed to them (Section 10.5). They are listed in the order the class proposed them, and cited by the short titles of the References.

The NotPetya attack on A.P. Moller-Maersk's terminals in June 2017 (DBA-TRAN-RE-1) is the first common-platform case. On 27 June 2017 destructive malware was delivered through the update system of a Ukrainian tax program, on the accounts of US-CERT and Talos; it entered the company through that program, on the company's account, and spread through its network, and the company shut down systems to contain it. Container terminals of its terminal company, APM Terminals, on five continents were closed or slowed: seventeen by the trade press's count, the company giving no count and the chairman of a port trust speaking of seventy-five. Pier 400 at the Port of Los Angeles was shut for three days and Maasvlakte II at Rotterdam heavily impacted for about a week; most terminals worked by hand at reduced productivity within two or three days, and all were reported open by 5 July 2017 (company statements of 28 and 30 June 2017; Port Strategy; Seatrade Maritime; FOX 11 Los Angeles). The company told its investors that it expected the attack to reduce its results by 200 to 300 million dollars and that volumes were affected for a couple of weeks in July (A.P. Moller-Maersk, *Interim Report Q2 2017*). Its chairman spoke, as reported, of 4,000 new servers, 45,000 new PCs and 2,500 applications in ten days. The United Kingdom and the United States attributed NotPetya to the Russian military on 15 February 2018, and Canada, Denmark, Australia and New Zealand spoke on 15 and 16 February in their own words.

The ransomware attack on the Port of Nagoya in July 2023 (DBA-TRAN-RE-2) is the second common-platform case, and the purest: its system was shared by every terminal of one port. At about 6:30 in the morning of 4 July 2023 the Nagoya United Terminal System, which runs the port's five container terminals and its central gate, was found stopped; that is when the stoppage was found, not when the system was infected. The system was restored on the morning of 6 July and relinked at 14:15, after malware had been found in the backups and on the servers restored on 5 July, and the terminals restarted between 15:00 and 18:15 on 6 July once the restored data had been checked against the yard. The gate was shut for about three days, and about 20,000 containers were affected on the estimate of the ministry's investigating committee. On the ships the two primary accounts differ: the committee records that ship-side handling continued by hand, thirty-seven ships being delayed by up to about twenty-four hours, while the Nagoya Harbor Transportation Association's report says that work stopped at all terminals; the six ships Lloyd's List reported waiting were counted after the restart. The port handled about 10.7 percent of Japan's trade in 2023 on official statistics, and Toyota's export-parts packing lines stopped while its finished-car production continued (*Nagoya Ransomware* F-8). Every crane and every berth was intact. The port could not move boxes through its gate, because the one system that knew where every box was could not be used.

The power outage at Hartsfield-Jackson Atlanta International Airport on 17 December 2017 (DBA-TRAN-RE-3) is the first power case. That afternoon a failure of Georgia Power's switchgear, in a tunnel beneath the airport's concourses, started a fire that damaged the cables of the airport's main and redundant supplies and the switch that accessed the redundant system, and shortly after one o'clock power was out throughout the airport, on the accounts of Georgia Power, the City and the airport. The control tower kept its power on a separate feed, on CBS News and the Associated Press's account, and Georgia Power's timeline, as reported, records power restored to the tower's feed and the North Airfield Lighting Vault at 8:05 the next morning. The Federal Aviation Administration stopped flights to the airport. Power returned to all essential activities by 11:45 that night and to the whole underground network on 22 December. CBS News and the Associated Press reported more than 1,500 flights cancelled, a figure they did not attribute and that other counts put lower; about 1,400 were the airport's largest airline's own count. The redundancy that failed was in the utility's equipment inside the airport's fence, whose design was the utility's on the account of the airport's general manager.

The closure of Heathrow Airport after the fire at the North Hyde substation in March 2025 (DBA-TRAN-RE-4) is the second power case. At 23:21 on 20 March 2025 a high-voltage bushing of a supergrid transformer at North Hyde failed, most likely through moisture, and the transformer caught fire; at 23:49 the fire took the adjacent transformer, and the standby transformer, on the same circuit, was lost with it. Every supply from the substation ceased, and 71,655 customers lost power, among them the airport's largest supply point. The network restored some 42,000 of those customers within the hour and the last early the next afternoon, and the airport's supply point at 09:49. The National Energy System Operator's review found that the airport has three supply points, each feeding a discrete part of it, so that the loss of one cuts power to some of its operationally critical systems; that the network companies were not aware of the airport's internal configuration or of the impact of losing one supply point; that network operators generally do not know which of their customers are critical national infrastructure; and that there is no explicit requirement across sectors for critical infrastructure to operate through a loss of power (*Final report from the review into the North Hyde Substation outage*, dated 30 June 2025). The airport's plan put the switchover at ten to twelve hours. Counted from the loss of supply, power reached the terminals in about eleven hours and the whole of the airport's network in about fourteen and a half; counted from the start of switching at 02:00 it took about nine and twelve and a half, which the review Heathrow commissioned found within the plan, on the operator's account. Heathrow was closed for most of 21 March. The airline operators' representative counted about 1,300 flights cancelled, and the Press Association some 270,000 passenger journeys cancelled, diverted or delayed over 21 and 22 March.

The bombings at Brussels Airport and Maalbeek metro station on 22 March 2016 (DBA-TRAN-RE-5) are the occupancy case. Two devices exploded in the departure hall at Zaventem at 07:58, a third at the airport not being set off, and a device exploded in a metro train at Maalbeek at 09:10 or 09:11. The Minister of the Interior decided at 08h52, between the two attacks, to close the metro, and the decision reached the operator at 09h23, on the first interim report of the

parliamentary inquiry commission. Thirty-two people were killed and 340 injured on the count the commission's summary gives a week after the attacks; the court later counted thirty-five dead, and the press counted sixteen at each node. The metro reopened by stations to 25 April. The airport reopened on a limited basis on 3 April, with three flights that day and temporary facilities able to handle about a fifth of its former capacity, partly reopened its departure hall on 1 May and was fully operational again in June (commission's summary; Agence France-Presse; PBS NewsHour); its traffic for 2016 was about seven percent below the year before (VRT). In effect the crowd in the airport's busiest public space was the target; no source establishes a reason for the attackers' choice.

The congestion of the San Pedro Bay ports from late 2020 to 2022 (DBA-TRAN-RE-6) is the throughput case without a single cause. As import demand surged and the trucking, chassis, rail and storage beyond the terminals could not keep pace, container ships queued off Los Angeles and Long Beach. The queue began to build in October 2020 and the anchorages were full at the end of January 2021, on the account of the Marine Exchange of Southern California, whose executive director put the normal number of container ships at anchor at between zero and one in a statement to Insider in July 2021, as Fresh Fruit Portal reported it; the Associated Press reported a handful. The queue peaked at 109 vessels on 9 January 2022, counted under the queuing process of November 2021 that held ships beyond 150 miles, and the Marine Exchange declared the backup over on 22 November 2022, twenty-five months after it began. Both ports moved record volumes in 2021. The waiting ships filled the anchorages, and on 25 January 2021 two of them dragged anchor across a pipeline, on the safety board's account. No node was lost. The event shows how slowly a concentration node recovers once its flow exceeds what it can clear, and how far the delay propagates.

The ransomware attack on Collins Aerospace's check-in and boarding system in September 2025 was proposed as the third common-platform case and conferred as DBA-TRAN-RE-7. On the author's triage ruling of 9 October 2026 it is withdrawn as a reference event, its shared-platform findings being held by the NotPetya and Nagoya instruments, and it is cited here as an illustration. The attack, detected on 19 September 2025, disabled the MUSE system that many airlines share at the gates and counters of a number of European airports. Heathrow, Brussels and Berlin Brandenburg reverted to manual check-in and boarding, with handwritten boarding passes at Brussels, and Brussels asked airlines to cancel half of their departures on the worst days, by press reporting; it returned to normal operation on 29 September, after about ten days (Travel Extra; VRT; BX1). Reporting of the daily cancellations differs between outlets, and nothing here rests on a daily figure.

Three further events are cited rather than proposed. The explosion in the Port of Beirut on 4 August 2020 was conferred in the chemical series for its release chain and is withdrawn on the author's triage ruling of 9 October 2026; its findings on material held in custody awaiting disposal are carried by DBA-CHEM-RE-5 (*The West Fertilizer Explosion*) F-11 and F-12, and F-11 notes the custody finding to this class (Section 4.4). For this class Beirut shows the node

consequence of a release at a port, where the country's only grain silos and much of the port's bulk-handling capacity were destroyed, a reading that is this class's own and is cited to its sources. The explosions at the Port of Tianjin in August 2015 and the global failure of a security software update on 19 July 2024, which grounded flights at many airports at once, are cited as illustrations.

The design-basis reading of the six is direct. At Nagoya and Maersk the nodes were physically whole and could not move at their design rate, because the system that knew what to move was gone: Nagoya could not move boxes through its gate, the committee recording ship-side handling by hand and the association's report a stop at all terminals (Section 2.3), and at Maersk most terminals moved by hand within days (*NotPetya Attack* F-4). At Atlanta the redundancy that failed was in the utility's equipment inside the fence. At Heathrow the airport's supplies were independent, and failed as redundancy because none could carry another's load in time. At Brussels the node held a crowd, and in effect the crowd was the target. And at San Pedro Bay the measure that mattered was not whether the node worked but how much of its flow could go nowhere else, and for how long. Those are the class's measures.

3. The Facility-Class Design Basis

3.1 Why Nodes Are the First Class

The node is where the sector's consequence is most concentrated and where its boundaries with other sectors are most crowded. Petroleum, gas and chemical terminals stand inside its ports; the electric series supplies power that, at a hub airport, no other source replaces quickly; the Commercial Facilities series builds venues over its stations; and the military deploys through its seaports. Writing it first fixes the boundaries that the ways, the conveyances and the control class will each meet, and lets the control class be written last against what all three require of it.

3.2 The Two Consequence Chains

Chain T — throughput. The node cannot move what it is meant to move. The cause may be the loss of its berths, runways, cranes or yard; the loss of its power; the loss of the systems that plan and record its moves; the loss of access by the ways that feed it; or flow that exceeds what it can clear. Its size is set by the flow that cannot be sent through any other node, and by how long the node takes to recover. San Pedro Bay and the two common-platform events are its reference cases, and Brussels engaged it as well, the metro for its line and the airport for its flow (*Brussels Bombings* F-6).

Chain O — occupancy. The people the node holds are harmed. The cause may be fire or smoke in a terminal or station, a crowd that cannot clear the space it is in, the failure of the node's ventilation, lighting or public address while it is full, or an attack on the crowd. Its size is set by the number of people at risk and by how long they take to reach safety. Brussels is its reference case in direction; it supplies no measured clearance time, which rests on exercises and on the codes the class cites (*Brussels Bombings* F-8).

On the author's decision of 8 October 2026 a node carries whichever of the two chains applies. A freight node carries Chain T. A transit station carries Chain O, and Chain T for its line. A hub airport and a passenger ferry terminal carry both, and must meet both at once: the event that stops an airport's flow also leaves tens of thousands of people inside it without power, information or a way home. At Atlanta the people inside, of about 30,000 affected on the mayor's figure, were held for about five hours without normal light, public address, escalators or screening before police, rail and buses cleared them (*Atlanta Power Outage* F-5). Heathrow tested Chain O at the node's lowest occupancy and its record gives no count of the people inside, so the occupancy reading rests on Atlanta (*Heathrow Closure* F-4).

The two chains pull against each other in one way. The quickest way to protect a crowd is often to close the node, and the quickest way to restore throughput is to reopen it. A design basis for a node that carries both must say who decides between them, and on what basis (Section 6.3).

3.3 The Governing Measures

Two measures govern the class, on the author's decision of 8 October 2026.

Stranded flow is the share of a node's design throughput that no other node can absorb within the time that matters, carried for as long as the node is down. It has two parts. The first is the node's recovery time: how long, after an event begins, until the node has regained its design throughput, or a stated fraction of it. The second is its diversion capacity: how much of its flow other nodes can take, how soon, and by what route. A node whose flow can be diverted strands little even when it is lost for weeks; a node with no absorbing alternative strands all of its flow for as long as it is down. Stranded flow is the node's counterpart of the long-lead recovery measure the energy series uses for equipment that cannot quickly be replaced (framework §3.3), and it is the measure the framework's sole-path elevation needs (framework §6.1). It is counted for each kind of flow the node carries, because a port that can divert its containers may have no other place to land automobiles or heavy machinery.

Stranded flow counts flow delayed as well as flow lost. At a node that keeps working but cannot clear its flow, it is counted as the container-days or vessel-days of waiting beyond the node's normal wait, offshore and on the terminal; in 2021 the San Pedro Bay ports moved record volumes while ships waited offshore for up to two weeks (*San Pedro Bay Congestion* F-1). A node that depends on a shared platform counts its stranded flow at each interface through which its flow passes, at a container terminal the ship and the gate at least, for each kind of flow, and after any loss of the platform records the flow stranded at each in measured units, so that its own sole-path calculations and its dependents' rest on measured values (*Nagoya Ransomware* F-8).

Recovery time runs to the return of the node's flow, not of its power or its systems. After the loss of a supply point it is the switchover time, plus a restart time, the time to check, restart and return to service the systems that lost power, which each node states and measures, plus the time for its flow to return, including the conveyances and crews its closure displaced. Heathrow took about twenty-eight hours from the loss of supply to full operation, and Atlanta's flow returned a day or

two behind its power (*Heathrow Closure* F-3). After the loss of a shared platform, recovery time includes the time the operator's isolation of the platform holds the nodes down and the time to restore the platform from sound copies and reconcile its records (Sections 4.2, 5.6, 6.3). After an attack on a passenger node it runs to the end of the security regime under which the node reopens, whose rate is stated: Brussels Airport reopened at about a fifth of its capacity, 800 departing passengers an hour on its own statement, and was fully operational only in June (*Brussels Bombings* F-6).

A diversion capacity is credited only where the information the diverting node needs, what each container is, where it goes and who may release it, reaches that node by a means that does not depend on the lost platform. In 2017 the carriers exchanged load lists and bay plans by other means so that other terminals could take the cargo (*NotPetya Attack* F-5).

Clearance time is how long the node takes to move its design occupancy to a place of safety. For a transit station it is set by the standard the station is built to, which for fixed-guideway transit and passenger rail is NFPA 130 (*Standard for Fixed Guideway Transit and Passenger Rail Systems*). For a passenger terminal it is set by the building and fire codes and by the terminal's own evacuation plan. The design occupancy is the largest crowd the node may hold, including the crowd from a venue or event that discharges into it, and not the crowd of an ordinary hour. Beside its clearance time each passenger node states its hold time: how long it may hold its design occupancy in place with its normal systems degraded before it must clear it, sized with its point-of-use supplies (Section 4.1). The occupants of conveyances held on the node's apron are counted in its design occupancy once they are brought off (*Atlanta Power Outage* F-5), and on the author's decision of 10 October 2026 the people aboard a train standing at a station's platform are counted within the station's occupancy.

The design basis requires that, for the events a node must withstand, its stranded flow stays within the limit set for it, and that its design occupancy can reach safety within its clearance time with the node's normal systems degraded.

3.4 Design Parameters at Class Resolution

Two further parameters are set at class resolution, on the author's decisions of 8 October 2026.

The manual fallback rate and hold time. Every node that depends on a shared software platform must state the rate at which it can move flow, check in passengers or board aircraft by hand when the platform is lost, and how long it can hold that rate before staff, paper and patience run out. The fallback rate is a design quantity, not an aspiration: it is measured in exercises and in real events, and a node that has never measured it does not know it. It is stated for each interface through which the node's flow passes and for each function the node performs, at a container terminal the ship and the gate at least, and the berth, yard, rail, imports and exports separately, with the hold time for each, because a node with a fallback at one interface and none at another strands the flow of the second for the whole of the platform's loss. Nagoya had no fallback at its gate, which stayed shut about three days, while the ministry's committee records that ship-side

handling continued by hand from printed work instructions, which the harbor transportation association's report does not mention (*Nagoya Ransomware* F-1). Across Maersk's terminals in 2017 the rates ran from nothing at Pier 400 for three days to reduced productivity elsewhere within the same days, with no hold time in the record against which the days of manual work can be read (*NotPetya Attack* F-4).

The rate is stated with the people who supply it. Nagoya's manual working was possible because people were present who had worked by hand before the system. A node trains and exercises its staff so that the rate does not rest on those who learned the work before the platform, states the decline of the rate as such staff leave, and does not credit a rate held by people it does not replace for longer than they remain (*Nagoya Ransomware* F-2). The hold time is reckoned with the replacement of the crews who would work by hand, as the framework counts campaign capacity (framework §7.6; *Strategic Seaport Denial*). A node also states its rate for flow whose records are suspect, and its rate under the security regime it would impose after an attack on it, each measured in exercises beside its rate on the loss of a shared platform (*Brussels Bombings* F-6). During the Collins Aerospace attack of September 2025 (Section 2.3) Brussels worked for about ten days on handwritten boarding passes.

The power switchover time. Every node must state, for the loss of any one of its supply points, how long its operationally critical systems are without power before they are restored from another, and must hold that time within the limit set for it. The time runs from the loss of the supply point to the restoration of power to the last operationally critical system, and it includes the time to decide to switch and to bring the people who switch. A switchover time that has not been exercised end to end is not known and is not credited (framework §4.2). Heathrow's plan said ten to twelve hours; counted from the loss of supply, power reached the terminals in about eleven hours and the whole network in about fourteen and a half, about two hours passing before switching began, and the scenario had never been exercised (*Heathrow Closure* F-2). The network companies were not aware of the airport's internal configuration or of the impact of losing one supply point (*Heathrow Closure* F-5). For the event Section 4.1 names, a supply lost with the equipment that transfers to its backup, the node also states its time without power where no transfer remains, as a time to restore by repair that includes the time to reach the damaged equipment through the fire, smoke and fumes the event leaves. At Atlanta that was about six hours to the first concourse, eleven to the concourses and nineteen to the tower's feed and the airfield lighting, nearly four of them before the utility could reach its equipment (*Atlanta Power Outage* F-2). The switchover time is to be known to the electric utility that serves the node (Section 4.1).

The measures on a queue. A node whose flow can exceed its clearing capacity states the measures it would take, among them extended hours, charges on dwell, and the order and place of the queue; the parties beyond its gate on whom each depends; and each measure's expected effect. A measure whose effect has not been measured is not credited. At San Pedro Bay

extended hours found very few takers, and a fall in aging cargo after a dwell fee announced and never collected cannot be separated from the easing of demand (*San Pedro Bay Congestion* F-7).

The values of these parameters, of the stranded-flow limit, the clearance time and the hold time, and of the restart time depend on the node and are not given here as numbers (Section 10.1).

4. Design Basis Events: Non-Adversarial Set

The framework lists candidate bounding events for the sector, among them the loss of a high-throughput node, but does not develop them (framework §4.1). This section develops the events for the nodes under the series' four headings: the event, the harm that makes it count, what must be done, and for how long. Section 5 applies the military-action extension to them.

4.1 Loss of Supporting Power

Postulated event loss of the electric supply at one or more of the node's supply points, through a fault, fire or failure at the utility's substation, on equipment another party runs inside the node's fence, or on the node's own distribution network, including a single event that takes both a primary supply and the equipment that transfers to its backup at the place where they come together.

Consequence threshold a node whose operationally critical systems — terminal lighting and life safety, baggage and cargo handling, cranes, gates, fuel distribution, its operating and information systems — are without power for longer than its switchover time allows.

Required function the node keeps its operationally critical systems running when it loses any single supply point, and states the time its systems are without power before another supply restores them (Section 3.4). Redundancy of supply is shown by the arrangement as built and not by the number of supplies, and by two conditions together. First, the supplies share nothing whose loss defeats more than one. A backup that shares a room, a duct, a switch or a fire zone with the supply it backs up is not independent of it; the test applies along the whole route of each supply from the supply point to the critical load, including any length of it that another party owns inside the node's fence, and the place nearest the load where two routes, or a route and the transfer between them, come together is shown separately. Atlanta took its power from two separate substations, and one fire took the cables of both and the switch to the redundant system where they ran together (*Atlanta Power Outage* F-1). Second, the critical loads of any one supply point can be carried by the others within the switchover time, by automatic or remotely operated transfer or by a switching plan whose time is measured. At Heathrow the three supply points shared nothing the fire reached and two kept their power, but each fed a discrete part of the airport and moving the lost one's load took manual switching through most of the following day (*Heathrow Closure* F-1). The node's life-safety systems, namely emergency lighting, public address, smoke control, the means of moving people who cannot use stairs or stopped escalators, and the systems that open the way out, run on supplies at the point of use. Those supplies, with the supplies of the systems needed to hold occupants safely in place, last the longer of the

clearance time and the switchover time, so that a node that cannot clear within its backup's endurance does not rely on that backup through a switchover. At Atlanta the generators lit the emergency lighting, the public address could not be used and disabled passengers were carried; Heathrow's own answer of 2018 was that its backed-up systems would meet immediate safety concerns but not continue operations (*Atlanta Power Outage* F-4; *Heathrow Closure* F-4).

On the author's decision of 8 October 2026, the node's power arrangements, its critical loads and its switchover time are made known to the utility that serves it, so that the utility knows which of its customers cannot ride through the loss of a supply point; Heathrow showed that otherwise they are not known. They are made known by a written exchange naming the supply points, the critical loads on each, the switchover time and what each party means by a resilient supply, renewed on change, and the knowledge runs both ways: the node knows the routes and switching of any supply another party runs inside its fence, as the utility knows the node's critical loads (*Heathrow Closure* F-5; *Atlanta Power Outage* F-3). On the author's decision of 10 October 2026, the node states what it requires of a supply another party runs inside its fence, namely its routes, its independence shown as built and its times, and the electric series designs that equipment. The requirement reaches the utility as quality of service, through an agreement or tariff that names the event the node's supply must survive. A statistical service level does not discharge it, because how often and how long customers lose supply is a measure of likelihood, and the inclusion criterion is consequence. Nor does the exchange buy priority: the network companies' licenses generally prohibit discrimination between customers in restoration, and Heathrow's lost supply point came back when the fire allowed (*Heathrow Closure* F-5). The exchange does not relieve the node of riding through the loss of any single supply point on its own (framework §4.1). Section 8.3 and Section 10.3 carry the matter to the electric series.

Design-basis duration the switchover time for the critical systems, or where no transfer remains the time to restore by repair, and for life-safety systems the longer of the clearance time and the switchover time.

4.2 Loss of a Shared Software Platform

Postulated event failure of a software platform that many nodes share — a terminal operating system, a common-use check-in and boarding system, a baggage or cargo system, the endpoint protection or operating system on the node's computers — through a faulty update, a supplier's outage or the loss of a cloud service, so that every node running it fails at once.

Consequence threshold a node that is physically intact and cannot move flow, check in passengers or board aircraft because the platform that plans and records its moves is lost.

Required function on the author's decision of 8 October 2026 this is a design-basis event of its own, and it is the class's common-cause event. Each node states the shared platforms it depends on, which other nodes depend on the same ones, and its manual fallback rate and hold time for the loss of each (Section 3.4). The node holds, in a form that does not depend on the platform, what it needs to work by hand: the position and contents of what is in its yard, the booking and

loading lists for what is due to leave, and the procedures and forms to process passengers. It exercises the fallback at a stated interval, with the platform assumed lost. An update that can reach the platform at every node at once is not applied to all of them in one operation without a governed act, meaning an authorized and recorded change made for each group of nodes the change could take out together (*Local Sovereignty* §3.2). The global failure of a security software update on 19 July 2024, which grounded flights at many airports in one morning, is the non-malicious case.

The shared platforms a node states include the operator's enterprise network and identity domain wherever the node's operating systems depend on them, and the software of the operator's other businesses and offices that shares that network. The common-cause unit is everything one compromised credential or one exploitable service can reach, so that a node whose terminal operating system shares a domain with an office in another country shares that office's exposure. Maersk's terminals stopped because the malware entered the company's network through a tax program at one office in Ukraine and spread to every terminal by stolen credentials and file-sharing exploits (*NotPetya Attack* F-2). Where a platform is shared by several operators at one node, or by several nodes, the node's plan names the party that owns and maintains it and records, by contract between that owner and the platform's users, who holds each of the platform's requirements under this section and Section 4.8, among them its backups, its external access, its fallback procedures and their exercise. A node made of several terminals on one platform is one node for this section. Nagoya ran one system for five terminals and its gate where Tokyo ran eleven, and the system's users and its owner differed (*Nagoya Ransomware* F-3).

The shared platforms on which a node depends, the operator's identity domain among them, are restorable from copies that no system an attacker could reach can alter or destroy, within a stated restoration time that is exercised end to end and counted in the node's recovery time (Section 3.3), as the framework requires that sound software can be restored to every unit quickly (framework §4.3). On one account Maersk's rebuild rested on a single copy of its identity domain that a power cut had kept offline (*NotPetya Attack* F-6). A backup counts as independent of the platform only where it reaches back beyond the longest time an intrusion may go undetected, in several generations, is kept apart from the platform, covers every environment needed to rebuild it, and is shown clean before it is restored. Nagoya held three days of backups of its production environment alone; malware was found in them and on the servers restored from them, and no clean copy from before the infection existed (*Nagoya Ransomware* F-4).

Design-basis duration until the platform is restored from sound copies, the node's records are reconciled with what it holds, and the backlog is cleared. At Nagoya the terminals restarted after about two and a half days; Maersk's terminals were reported open within eight days, and volumes were affected for weeks.

4.3 Natural-Hazard Exceedance

Postulated event a hurricane, storm surge, flood, earthquake, wildfire or extreme heat that exceeds the node's design for the hazard, damaging its berths, cranes, runways, yards or stations, or flooding its electrical and systems rooms.

Consequence threshold a node whose recovery time, for the hazard the region must be designed for, leaves stranded flow above its limit; and a passenger node whose occupants are exposed while it is full.

Required function each node states its exposure to the region's design hazards, and the electrical rooms, switchgear, systems rooms and backup power on which its operation depends are placed or protected so that the hazard that damages the node's structures does not also take its power and systems. Cranes are secured against the design wind. A node whose flow cannot be diverted carries a recovery plan with the spares, contractors and equipment it would need, stated against the region's design hazard and not against the last one.

Design-basis duration the node's recovery time for the design hazard.

4.4 Fire, Explosion or Release of Cargo at the Node

Postulated event a fire, explosion or toxic release of hazardous cargo stored or in transit at the node — in a warehouse, a container stack, a tank farm or a rail car in the yard.

Consequence threshold loss of the node's berths, yards, warehouses or terminals, and harm to the people in and around it.

Required function under the release-and-network rule of Section 8.3, on the author's decision of 8 October 2026, the release and its harm to people are governed by the chemical or energy series that holds the material, and this class governs the loss of the node as a transport node. Both use the shared release model (*Chemical Production and Synthesis* §4). For this class the requirement is that hazardous cargo is not held where its release would take the node's critical functions with it: the node knows what hazardous cargo it holds and where, keeps it apart from its critical structures and systems and from the spaces that hold crowds, and does not let it accumulate beyond the quantities its plan allows. Beirut is the measure of the alternative: the explosion destroyed the country's only grain silos and cut the port's bulk-handling capacity to a fifth or less, at a port that had met most of the country's import needs (United Nations officials, as reported by Reuters, September 2020). Material held at a node awaiting disposal, in custody or under seizure, belongs to the design basis of the class that holds it, as DBA-CHEM-RE-5 (*The West Fertilizer Explosion*) F-11 finds from the Beirut record and notes to this class; the node knows what such material it holds and where, and holds it under the same separation as any hazardous cargo. At Tianjin in August 2015 the official investigation reported 165 dead and some 7,500 containers and more than 12,000 cars damaged.

Design-basis duration the node's recovery time after the event.

4.5 Loss of Access by the Ways

Postulated event closure of the channel, airspace, track or road by which flow reaches the node, through a collapse, grounding, wreck or closure beyond the node's fence.

Consequence threshold a node that is intact and cannot be reached.

Required function each node states the ways by which its flow arrives and leaves, which of them it cannot do without, and its stranded flow if each is closed. The design basis of the ways themselves is DBA-TRAN-WY-FC2's. The node holds what it can do inside its fence: the ability to send flow out by another mode or way, and to hold what is inside it safely while it waits. The collapse of the Francis Scott Key Bridge on 26 March 2024 closed the main channel to the Port of Baltimore, then the nation's leading port for automobiles and for roll-on, roll-off heavy machinery, until the Corps of Engineers restored it on 10 June, seventy-six days later (U.S. Army Corps of Engineers; Office of the Governor of Maryland). The way was lost; the node's flow was stranded. The event belongs to the ways class, which proposed it, and the framework conferred it as DBA-TRAN-RE-8.

Design-basis duration until access is restored, which at Baltimore was eleven weeks.

4.6 Flow Beyond Clearing Capacity

Postulated event flow arriving faster than the node can clear it, through a surge in demand, the loss of capacity at a neighboring node whose flow diverts to this one, a labor stoppage, or a backlog left by any of the events in this section.

Consequence threshold a queue that grows faster than the node can work it down, so that stranded flow, counted as flow delayed (Section 3.3), accumulates even though nothing is broken.

Required function each node states its clearing flow, the flow beyond which its queues grow without limit, and how long it takes to clear a backlog once the flow returns below that level. The clearing flow is the least of the node's berth rate, its yard capacity and the rate at which the chains beyond its gate take flow away, and the node states what it counts on of trucks, chassis, rail and storage. At San Pedro Bay the terminals moved record volumes, and what bound the complex was the rate at which containers left it (*San Pedro Bay Congestion* F-2). The dwell time of flow at the node is stated with its normal value, as the indication that the clearing limit is near. On the author's decision of 10 October 2026 the queue a node causes is this class's, as a consequence it measures, while the anchorages and holding areas where it waits are DBA-TRAN-WY-FC2's as places and DBA-TRAN-CntlCo's to manage. A node whose queue waits outside its fence states, with its clearing flow, the length of queue beyond which those places would be filled at the spacing their safe use requires; at San Pedro Bay the anchorages were full through 2021 and two ships dragged anchor across a pipeline (*San Pedro Bay Congestion* F-3). A node whose flow can exceed its clearing capacity for weeks or more carries, in a plan made with the control class and the parties at the node, the order in which waiting conveyances are

worked, the places where they wait and the holder who sets both, fixed before the event. At San Pedro Bay that arrangement was made nine months after the anchorages filled (*San Pedro Bay Congestion* F-4).

A node that is named as the diversion path for another node's flow, in that node's stranded-flow calculation, states how much it can take, at what cost to its own flow, and how it would receive the information the diverted flow needs (Section 3.3). A diversion capacity that is counted by more than one node is counted once. A surge of demand is common to every node that could absorb it, so the diversion credited against a surge is what the alternatives can take at their own surged flow, credited only on the recorded behavior of the importers and carriers who choose it; in July 2022 most of the container ships waiting off North America were off the East and Gulf coasts (*San Pedro Bay Congestion* F-5). San Pedro Bay is the reference: a backup that lasted twenty-five months, from October 2020 to November 2022, whose backlog fell from its peak to none in about ten and a half months as demand eased and flow moved to other coasts. A backlog that clears because demand falls or flow moves elsewhere has not been cleared by the node's capacity (*San Pedro Bay Congestion* F-6).

Design-basis duration the time to clear the design backlog, reckoned from the date flow falls below the clearing level to the date the wait returns to the node's normal. For a node whose flow cannot be diverted, the duration is set by the duration of the surge and not by the node.

4.7 Crowd Emergency at a Passenger Node

Postulated event fire or smoke in a terminal, concourse or station; the arrival of a crowd larger than the node can hold, as when a venue above or beside it discharges; a stampede or crush; the failure of ventilation, lighting, public address or vertical circulation while the node is full; or an attack on the crowd, whoever makes it.

Consequence threshold occupants who cannot reach safety within the node's clearance time.

Required function each passenger node states its design occupancy, including any crowd that a venue or event discharges into it, and its clearance time for that occupancy with one exit route lost. A station meets NFPA 130. The node's life-safety systems work without its normal power and without its information systems (Section 4.1). Where a station lies beneath a public-assembly venue, the station's plan and the venue's plan are made together and each states what it needs from the other, under the boundary of Section 8.3; the station's side is stated here. Beside its clearance time it states its hold time (Section 3.3).

On the author's decision of 10 October 2026 the attack on the crowd is a design-basis event of this section, because inclusion in the design envelope turns on consequence and not on who the attacker is; Brussels realized it in non-state form. Each passenger node states where its processes gather crowds outside protected areas and how large they grow, and holds them below the size its plan allows by its processes and its design rather than by its security posture alone. The requirement binds its security measures as well: a measure that moves the screening point outward states the crowd it forms, where and for how long, and holds it below the plan's size

before it is relied on. After the attacks, screening before the building at Brussels formed queues of two to three hours outside the terminal, which the director general of the airline industry's association called large and dense crowds that could be targeted (*Brussels Bombings* F-1). The node's clearance time is met with the area of the attack lost, and its life-safety systems keep working when its information systems and power are struck in the same act.

An attack on a passenger node is a design-basis warning to the passenger nodes that share its region, its crowds or its flows. Each such node states and measures its notification time, from an outside authority's decision to close it to the start of its own clearance, and holds the sum of its notification time and its clearance time within the interval its plan allows; the decision reaches the node's operator by a direct, acknowledged channel from its holder (Section 7.3). At Brussels the decision to close the metro was taken at 08h52 and reached the operator at 09h23, and the device at Maalbeek exploded at 09:10 or 09:11 (*Brussels Bombings* F-2). The passenger node's emergency plan holds the attack case, including the access and egress of responders and of the injured through the node's roads and spaces while it is clearing, and the information the node supplies to responders is marked as unverified until it is confirmed (*Brussels Bombings* F-4). The response is DBA-ESS-RF-FC2's (*Response Forces and Stations*), to which the point is noted.

Design-basis duration the clearance time, with the notification time before it where an outside decision closes the node, and the time to hold the node clear until it is safe to return.

4.8 Cyber Attack on Node Systems

Postulated event compromise of the systems that run the node — terminal operating systems, crane and yard automation, gate and access control, baggage and cargo handling, common-use passenger systems, building management, and the fire and life-safety systems — including through the suppliers and remote-access channels that maintain them.

Consequence threshold a single compromise that stops the node, that sends cargo, baggage or vehicles to the wrong place, that opens access the node should hold closed, or that defeats a life-safety system while the node is full.

Required function the loss of the node's information systems does not stop its safety functions, and the loss of either does not stop the node at a rate below its manual fallback rate (Section 3.4). Systems that can move equipment — cranes, automated vehicles, conveyors, doors and gates — are separated from systems that only plan and record, and a command from the planning side cannot drive equipment past the limits its own controls enforce. The records on which the node relies to know what it holds are protected against alteration, because a node that trusts a falsified record ships or stores what it should not. Ports subject to the Maritime Transportation Security Act meet the Coast Guard's cybersecurity rule (33 CFR Part 101, Subpart F); airports and railroads meet the Transportation Security Administration's requirements for their mode. The systems that run the node, its terminal operating system, its gates and its equipment controls, are separated from the operator's enterprise network and identity domain, so that their loss does not follow from that network's loss, and the separation is shown as built (framework §4.2; *NotPetya*

Attack F-2). A remote-access channel into the node's systems or a platform it shares is restricted to known sources, authenticated by more than one factor, patched on a stated interval, logged outside the system it reaches, and open only while in use. On the committee's reading Nagoya's attacker entered by a maintenance channel that accepted any internet address with a matching identifier and password, so that it could be reached quickly in an emergency (*Nagoya Ransomware* F-5). A remotely operated transfer of power between supply points is among the node's systems and carries these requirements (Section 5.6).

Design-basis duration indefinite; the condition lasts until the compromise is found and removed.

4.9 Loss of Communications and Timing

Postulated event loss of the communications networks the node uses to reach its tenants, carriers, customs and the control class, and loss or corruption of the satellite positioning and timing on which its automated equipment, its records and its systems depend.

Consequence threshold a node whose equipment stops or misplaces what it moves, or whose systems disagree about time, when communications or timing are lost.

Required function each node states which of its functions depend on satellite positioning or timing, and the tolerance of each. Its timing receivers meet the design basis held in *Positioning, Navigation and Timing* §7, which is not restated here. Automated equipment that positions itself by satellite stops safely when its positioning is lost or implausible, and does not move on positions it cannot verify. The node keeps working with its outside links lost at its manual fallback rate.

Design-basis duration the holdover time its timing receivers are designed to, and for communications the time to restore the failed link.

Table 4-1. Class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Measure
Supporting power	Loss of a supply point, or of a supply and its backup together	Chain T, Chain O	Power switchover time; restart time; time to restore by repair
Shared software platform	One platform fails at many nodes at once	Chain T	Manual fallback rate and hold time
Natural hazard	Hurricane, surge, flood, earthquake beyond design	Chain T, Chain O	Stranded flow

Event category	Postulated event (class-specific)	Consequence chain	Measure
Cargo fire, explosion or release	Hazardous cargo at the node burns, explodes or is released	Chain T, Chain O	Stranded flow; release chain cited
Loss of access	Channel, airspace, track or road to the node closed	Chain T	Stranded flow
Flow beyond clearing capacity	Queue grows faster than the node can clear it	Chain T	Stranded flow, counted as flow delayed
Crowd emergency	Fire, smoke, crush, an attack on the crowd, or loss of systems while full	Chain O	Clearance time; hold time; notification time
Cyber attack	Compromise of node systems	Chain T or O	Manual fallback rate; stranded flow
Communications and timing	Links or satellite timing lost or corrupted	Chain T	Manual fallback rate; holdover time

5. Military-Action Extension: DBA-MA for the Nodes

5.1 MA Extension Applicability Determination

Every node of consequence performs the facility-level applicability determination the series makes mandatory (framework §7), under the four-filter applicability gate of DBA-MA-AF (*MA Applicability Framework*) §2 and §3. Any one filter satisfied brings the node into scope. On the author's decision of 8 October 2026, the determination is presumed positive for two kinds of node. The first is the designated strategic seaport: the commercial ports of the National Port Readiness Network through which the armed forces deploy, of which eighteen are listed (Maritime Administration, *Commercial Strategic Seaports*), together with the military ports of embarkation, which this class does not govern (Section 8.3). The second is a node whose flow has no absorbing alternative at the scale of the nation or a region: a node whose stranded flow, if it were lost, would approach all of its flow. The San Pedro Bay ports, the Chicago rail hub and the express carriers' cargo hubs are of this kind, and so, on the author's decision of 10 October 2026, is a hub airport whose flow cannot be diverted at scale. Every other node runs the ordinary determination. A presumption can be rebutted only on a recorded basis (Section 5.8).

5.2 The Four Characteristics Applied to the Class

Applied to the nodes, the four military-action characteristics take specific form. Intentionality: the adversary chooses the node whose loss strands the most flow, or holds the most people, or

delays a deployment the longest, and times the act to the moment it matters most. Persistence: the adversary sits inside the platforms that many nodes share, or inside a supplier that maintains them, long enough to learn them and to reach every node at once. Precision: the adversary takes the one supply point, the one platform, the one berth or the one concourse whose loss stops the node, and leaves the rest alone. Denial: the adversary takes the power, the platform and the access together, and attacks the diversion path as well, so that the flow has nowhere to go. The class validation standard is the state actor who knows that the nodes are where the sector's flow, and the nation's ability to deploy, can be stopped with the fewest acts. Four further forms follow from the class's events. A node may be reached as the collateral of an attack aimed elsewhere: on the governments' accounts NotPetya's targets were Ukrainian, and the terminals stopped because the company's shared platforms joined them to an office in Ukraine (*NotPetya Attack* F-8). Denial at a passenger node includes the attack on the response and on the next node during the interval the first attack opens (*Brussels Bombings* F-7). Precision at a node near its clearing limit is a small cut in its clearing rate, at the constraint beyond its gate, which strands disproportionately while the surge lasts (*San Pedro Bay Congestion* F-8). And an act made to look like a failure of equipment is distinguished from one only by an investigation of cause, and would be read as a failure for as long as the cause took to establish. At Atlanta the question of foul play was answered in two days, and the record has not established the cause of the switchgear's failure; at Heathrow the cause took more than three months (*Atlanta Power Outage* F-8; *Heathrow Closure* F-8).

5.3 The Node in Its Military-Mobility Role: The Governing MA Scenario

The governing scenario is the denial of a strategic seaport during a deployment. Forces moving to a theater pass through a small number of designated ports, on a schedule that an adversary can estimate from public signs of mobilization. A coordinated campaign takes the port's terminal operating system and its gates by cyber attack, cuts its power at the supply points it depends on, and closes or threatens its channel, at the moment the first sealift ships are due. The port's commercial flow stops with its military flow, and the commercial ports that would absorb the diverted flow are struck in the same campaign or are already full.

The scenario turns on two things the node controls and two it does not. It controls whether it can move by hand at a known rate when its platform is lost, and whether its critical systems survive the loss of a supply point. It does not control its channel, which is the ways class's, or the diversion capacity of its neighbors. The design basis therefore requires, for a node whose determination is positive, that its manual fallback rate and hold time be stated for the military-mobility flow as well as the commercial flow, with the platform and the outside links lost together; that its power switchover time be met with the supply points an adversary would choose lost, not only the one most likely to fail; and that its plan name the ports that would take its military flow, checked against the same campaign. The supply point an adversary would choose is the one whose loss takes the longest to switch over, and any two where an adversary could reach two. It may also be the place inside the fence where supplies and the transfer

between them converge, rather than a supply point, and the switchover time is met with that place lost (*Heathrow Closure* F-8; *Atlanta Power Outage* F-8). The scenario is run with the node and its diversion ports at surged flow as well as at normal flow, because for two years San Pedro Bay's alternatives were full without any act (*San Pedro Bay Congestion* F-8).

The scenario sets the class's adequacy test. A node passes if, for the design duration of the campaign, it keeps moving its military-mobility flow at its stated fallback rate with its platform, its outside links and a supply point lost. A node that cannot pass carries a declared residual (Section 5.7).

5.4 The Crowd as the Target

The second scenario uses Chain O. On the author's decision of 10 October 2026 the attack on a crowd is a design-basis event of Section 4.7, and this extension keeps its compound form by a state actor. The adversary strikes several passenger nodes of several modes in one region in sequence, where and when each is fullest: the departure hall before a morning wave, a station platform at the end of an event, a concourse where a security queue has gathered a crowd outside the screened area. It strikes the response and the next node during the interval the first attack opens, and strikes the nodes' information systems and power in the same campaign. Brussels realized the sequence in non-state form, at two nodes of two modes about seventy minutes apart, and is the reference for its direction. On the author's ruling of 10 October 2026 a non-state attack that realizes the scenario carries no tier of its own, and the register describes it as adversarial, non-state and realizing the scenario (*Brussels Bombings* F-7). The design basis requires that each passenger node meet the requirements of Section 4.7 with the warning interval shortened to what such a campaign allows.

5.5 The Shared Platform as the Target

The third scenario uses Chain T, across many nodes at once. The adversary reaches the platform that many nodes share, or the supplier that maintains it, and at a chosen moment disables it everywhere it runs, or alters what it records. NotPetya is the reference for the path: an attack that governments attributed to a state, spread through a software supplier's update channel and through one operator's network, and that reached terminals on five continents on 27 June 2017. On the governments' accounts the nodes were not its target (*NotPetya Attack* F-8). The operator's decision to isolate its network to contain such an attack is a decision an adversary will reach for (Section 6.3). Alteration is the worse case, and no event of the class yet shows it: NotPetya destroyed what it reached and did not alter it, and it did not test the limits of the node's equipment (*NotPetya Attack* F-7). A platform that will not start is obvious. A yard plan that shows a container where it is not, a manifest that hides what a container holds, or a queue list or berth allocation that orders the wrong ships, is believed. The defense an altering adversary would aim to defeat is the reconciliation of the node's records with its physical stock before work resumes, which Nagoya performed terminal by terminal before restarting (*Nagoya Ransomware* F-7). An attack that presents as criminal ransomware is read as such for as long as no one looks

further, so the reading of an attack on a shared platform is revisited before the node relies on it (*Nagoya Ransomware* F-5). The class answer is the requirement of Sections 4.2 and 4.8, that every node can work by hand at a known rate and that an update that reaches every node at once is a governed act, together with the requirement that the indications on which any authorization rests, the queue list and berth allocation among them, satisfy *Indication Integrity*, so that an action authorized on falsified records is not taken as if it were sound (Section 6.3).

5.6 The Adequacy Result: Sustained Node Condition

The test of whether a node's protection is adequate is whether it can reach what this document calls the Sustained Node Condition. When the shared platform is lost, a supply point has failed, outside links are degraded and the diversion path is uncertain, the node still moves its flow at or above its manual fallback rate for the hold time it has stated; its operationally critical systems are powered within its switchover time; its occupants can reach safety within its clearance time; the records on which it relies are known to be sound or known to be suspect; and this holds for the design duration of the event or campaign.

The reconciliation of restored or suspect records with the physical stock is part of the node's recovery time. Each node states how it reconciles each kind of record and how long the reconciliation takes after a loss of the platform lasting its design duration, a time that grows with the age of the last sound record and with the flow moved by hand meanwhile. Nagoya's terminals restarted one by one as the check allowed (*Nagoya Ransomware* F-7; *Strategic Seaport Denial*).

The protective functions on which this condition rests are to be locally sovereign as FD-LS defines the property (*Local Sovereignty* §3.2). At the node they are the automatic transfer of critical loads to backup power, and between supply points, on a loss of voltage measured at the node, and the start of the backup generators, among them the point-of-use generation that carries the public address and the means of moving people who cannot use stairs, when a supply point is lost; the fire detection, suppression and smoke-control systems of terminals and stations, and the emergency ventilation of underground stations; the release of the doors and gates on the way out when the fire alarm operates; and the limits enforced by the controls of cranes and automated equipment — overload, overtravel, collision and wind — which stop the equipment whatever the planning system commands. Each decision is a measurement at the node — a loss of voltage, smoke or heat, a load, a position, a wind speed — against a setpoint; the function is colocated with what it protects; no system an attacker could reach can stop, delay or trigger its action; and no such system can alter its setpoints, logic or bypass state except through a governed act. *Local Sovereignty* governs and is not restated here. A smoke-control system whose modes can be changed over a building network an attacker can reach, or a crane limit that the terminal operating system can override, fails that test. Local sovereignty describes what protects the node and its occupants when the systems around them have failed. It does not decide whether any action needs human authorization (*Local Sovereignty* §4). A transfer between supply points that is operated remotely is not locally sovereign, and carries the requirements of Section 4.8

(*Heathrow Closure* F-1; *Atlanta Power Outage* F-4). The locally sovereign transfer depends on a passive protection, the physical separation of supply routes and transfer equipment from each other and from a common fire (*Atlanta Power Outage* F-8), and the point-of-use supplies of life-safety systems are sized to the longer of the clearance time and the switchover time (Section 4.1).

Each node must also hold its critical and life-safety functions for a set period with no outside power, communications or support. That requirement is written here in this document's own words. Other instruments take a similar requirement from the DBA-MA parent framework's "local sovereignty principle", but whether that principle is the same thing FD-LS defines is still an open question, so this document does not borrow it (Section 10.4).

5.7 Adequacy and the Declared Residual

In the non-adversarial domain, the Sustained Node Condition is achievable with the Section 4 requirements: power arrangements whose independence is shown as built and known to the utility, a manual fallback measured and exercised, hazardous cargo kept apart from the node's critical functions, a diversion path that is counted once, and life-safety systems that work on their own. Against a deliberate attacker, adequacy has limits.

Three residuals are declared, in the output form of DBA-MA-DiD (*Defense in Depth*) §3.5. First, a node that falls back to working by hand moves less than it did, and in a campaign the difference accumulates as stranded flow that the node cannot recover while the campaign lasts. The residual is that accumulated flow, and the dependent facilities in other sectors that it reaches; it is bounded by the node's fallback rate and by the stocks those facilities hold, and the decision to accept it is the responsible authority's, recorded in the node's plan. Second, where an adversary closes a node's access and strikes its diversion path in the same campaign, no measure inside the node's fence restores its flow. The residual is the stranded flow of the node and its neighbors together, and the answer to it lies with the ways class, the control class and the national response, not with this class. Third, on the author's decision of 10 October 2026, a residual is declared for the people a passenger node holds. Where an adversary strikes several passenger nodes of a region in sequence (Section 5.4) and the interval between one attack and the next is shorter than a node's notification time and clearance time together, no measure inside the node's fence clears its crowd before it is struck. The residual is the people the node holds in that interval, bounded by its design occupancy and by the clearance it has achieved; the answer to it lies with the authorities that may close the node, the responding agencies and the national response, and the decision to accept it is the responsible authority's, recorded in the node's plan (*Brussels Bombings* F-7). These residuals bound the class adequacy claim.

5.8 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter — a diversion capacity credited to a neighboring node that the same event or campaign would strike or that another node already counts, a backup supply credited without showing it independent of the supply it backs up, a manual fallback rate that has not been measured, a design occupancy set from an ordinary hour

rather than the largest crowd the node may hold, a supply credited as continuous on a judgment of its likelihood, which the framework's inclusion criterion, set by consequence, does not allow (framework §4.1; *Heathrow Closure* F-8), a remote-access channel opened for speed of response without a recorded basis (*Nagoya Ransomware* F-5), a security measure adopted after an event without a statement of the crowd it forms (*Brussels Bombings* F-1), a diversion capacity credited on behavior of importers and carriers that has not been recorded (*San Pedro Bay Congestion* F-5), a presumption of applicability rebutted without a recorded basis — must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework adopts this principle for the sector (framework §4.2).

6. The Dual Design Basis, Stated in Full

6.1 The Two Design Bases

A node carries two design bases at once: a physical design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). For passenger screening some actions carry a third consequence, governed by a second Line (Section 6.4). The bases interact at the point developed in Section 6.5.

6.2 The Physical Design Basis: The Determinism Posture

The physical design basis is the set of protective functions the node relies on to keep its critical systems powered, to protect its occupants, and to keep its equipment within safe limits, independently of the systems an attacker could reach. These are the transfer to backup power, the fire, smoke-control and emergency ventilation systems, the release of the way out, and the limits enforced by the controls of cranes and automated equipment (Section 5.6).

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Power transfer, smoke control and equipment limits that behave predictably within a stated range of inputs, and that have been formally tested as installed, are exempt from the Command Broker requirement for the actions they take while their inputs stay in that range (*Layer Decomposition and Determinism* §4.3). They are not thereby placed on either side of the Bright Line: a smoke-control system decides whether a crowd can breathe on its way out, and it is outside the obligation because its behavior is bounded by proof (*The Bright Line* §4.5). A component that has not been shown to meet the test is covered by the rules, and each of its actions is judged on its consequence.

6.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems are arriving at the nodes in several forms: automated container terminals, in which stacking cranes and guided vehicles are driven by the terminal operating system; optimizers that plan berths, yards and stowage; airport systems that

assign gates and stands and sequence turnarounds; systems that read crowd density from cameras and direct passenger flow, open and close gates, and change signs; systems that screen passengers and baggage; and assistants that help operators read alarms and decide what to do in an emergency. They are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-Framework_Cross_Sector (*Command Broker Framework*). CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. This document names the actions at the nodes that the architecture applies to. Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3).

Four cases illustrate the application, and none of them is a placement by component. A stowage or yard plan that places incompatible hazardous containers together, or places them beside a node's critical structures or crowds, is above the line where the release it could cause reaches the threshold; the same optimizer's plan for ordinary containers is placed on its own consequence. A move of an automated crane or vehicle into a space where people may be is above the line where a collision could kill; the controls' own limits stop it regardless (Section 5.6), and the authorization obligation reaches the command that sends it. A crowd-management system that closes a gate, stops an escalator or directs a crowd toward an exit, when the node is full, is above the line, because the action it takes decides whether people can leave. And a decision to close a passenger node, or to reopen it, is an action whose consequence falls on both chains at once (Section 3.2): closing protects the crowd and strands the flow; reopening restores the flow and returns the crowd to the risk. This document places that decision with the node's operator, as the holder of the node's Command Broker role, on the basis the control class and the responding agencies provide, and the control class, when it is written, states what it supplies to that basis (Section 10.9); a system that recommends it carries the obligation of *Indication Integrity*, and a system that executes it carries the authorization obligation.

The decision to close includes the decision to hold a full node's occupants in place or to clear them, which at Atlanta several parties took over about five hours (*Atlanta Power Outage* F-5). The operator names in advance, in an authorization map of its own of the kind the coordination class keeps (framework §5.5; *Traffic Management, Signaling and Control* §3.3, §6.2), the holders of the decisions to close, to hold or clear, and to reopen, and the order in which each passes when a holder cannot be reached. At Heathrow the chief operating officer decided at 01:11 to close the airport on the impact on its operationally critical systems, the chief executive not being reachable that night (*Heathrow Closure* F-6). Where a node's supply is partitioned it may reopen by part, each part on the state of the supply point and systems serving it, with the control class supplying the basis for movement (*Traffic Management, Signaling and Control* §6.3); Heathrow's Terminal 5 opened first. Where a public authority may close the passenger node by law, both act and neither waits on the other, as the coordination class provides for the same space (*Traffic Management, Signaling and Control* §6.3), and the operator's plan names each such authority, the channel by which its decision arrives and what the operator does on

receipt. At Brussels the metro's operator set off its own plan at 09h12, before the federal decision reached it at 09h23 (*Brussels Bombings* F-3).

Two decisions of the same kind belong to the loss of a shared platform. The isolation of a shared platform, or of the network on which many nodes depend, is a decision whose consequence falls on the stranded flow of every node it reaches. The operator names its holders and their succession in advance, plans the isolation to take out the fewest nodes the containment allows, by the separation of Section 4.8, and counts the time it holds the nodes down in their recovery time (*NotPetya Attack* F-3). And a node's plan for the loss of a shared platform to attack names in advance who decides between restoring the platform and preserving it for investigation, on whose advice, a cybersecurity specialist's among it, and what evidence is preserved first. The decision to resume work on a restored platform is a decision to reopen the node, made on a check that the platform is clean. At Nagoya restoration was given priority at about 10:30 on the first morning, the record not saying by whom; logs and data were lost through the restoration, and the servers restored from backup were found to carry malware (*Nagoya Ransomware* F-6; *Strategic Seaport Denial*).

Whether the obligation reaches an action depends on how the system is built. Where an operator reads a system's output and then decides and acts, the action is the operator's, and the output carries the obligation of *Indication Integrity* rather than the authorization obligation (*Mode of Discharge* §7). Where the system's output itself makes the change — a crane move that executes when it is planned, a gate that closes when the model says so — the obligation reaches it (*The Bright Line* §4.3; *Mode of Discharge* §7). A supplier must declare, for each output, which of the two it is.

Many of a node's automated actions are repeated thousands of times a day within limits set in advance, and the obligation for them may be discharged in advance under *Mode of Discharge*, within a pre-approved envelope (*Pre-Approved Action Envelope* §4) whose limits the node's operator sets and whose gatekeeper is verified. A change to those limits is a change to the envelope and is excluded from advance discharge (*Mode of Discharge* §4.3). The movement of anything beyond the fence — a vessel leaving its berth, an aircraft pushing back, a train leaving the yard — is an act of movement authority and is the coordination class's, which DBA-TRAN-CntlCo will place; the node supplies the basis on which that authority is given.

Consistent with CDB-6 of the Military Action Communications Design Basis, the loss or degradation of the node's own communications does not suspend the Command Broker obligation. CDB-6 states the obligation for actions recommended by generative AI; this instrument applies it to every component within the scope of *Layer Decomposition and Determinism* §4.3, which is the scope the Foundational Definitions set.

6.4 Two Lines for Passenger Screening

Screening a passenger carries two consequences. A system that clears or refers a passenger or a bag decides whether a threat reaches the aircraft, the train or the crowd beyond the checkpoint,

which is a consequence this document places under FD-BL. It also decides who is stopped, searched or held, and, where it matches faces or reads identity documents, whose identity is checked against what, which is a consequence on liberty that FD-SBL governs.

This instrument places each screening action on its security consequence, under FD-BL, and applies the obligation that placement creates. The liberty consequence of the same action is governed by *The Surveillance Bright Line*, whose placement criterion, scope gate and obligation are not restated here. As FD-SBL requires (*The Surveillance Bright Line* §2.1), this instrument names FD-BL for the security consequence and FD-SBL for the liberty consequence, and an action above either Line takes that Line's obligation; an action above both takes both. FD-SBL does not carry the determinism carve-out of *Layer Decomposition and Determinism* §4.3, so a deterministic screening system that is outside this document's obligation may still be within FD-SBL's (*The Surveillance Bright Line* §4.4). The same reading was made for law-enforcement actions in the Emergency Services series (*Response Forces and Stations* §6.4), and Project 1 is told in the same act.

6.5 The Interaction Point: Moving Under Denial

The two design bases meet when the node must keep moving while under attack. The control system must be architecturally incapable of executing an above-line action originating from a component within the scope of *Layer Decomposition and Determinism* §4.3 without Command Broker authorization, given in the moment or in advance under Section 6.3. That includes any hazardous stowage whose release consequence reaches the threshold, any equipment move into a space people may occupy, and any crowd action taken while the node is full. And the node's power transfer, life-safety systems and equipment limits must keep working even when no outside system or authority can be reached at all. The physical design basis guarantees that the node keeps its critical systems, its occupants and its equipment safe; the AI-governance design basis guarantees that no planning or crowd system can set up a release, drive equipment into people, or trap a crowd, without a qualified human authorization the architecture can enforce.

6.6 Scope Boundary

The physical design basis is defined in full in this instrument. The AI-governance design basis is named and referenced, not redefined; its governing text is *The Bright Line* with its determinations, the Command Broker family under its register, and, for the liberty consequence of screening, *The Surveillance Bright Line*. Where they are cited together, the physical requirement governs the node's protective architecture and the AI-governance requirement governs the authorization of any planning, crowd or screening system that interacts with it.

7. The Communications Design Basis Applied to the Nodes

7.1 Why It Applies

A node is a meeting place of many parties — the operator, its tenants, the carriers, customs and border agencies, the control class — and it works only as long as they can tell each other what is arriving, what is leaving and what is where. The Military Action Communications Design Basis applies to those links as it applies to any facility (*Military Action Communications Design Basis* §3). This section states how each of the seven requirements applies to the nodes. It cites them; their text is in that document.

7.2 The Finding That Shapes the Rest: A Node's Fallback Is a Rate, Not a Switch

A node does not fail over to a backup and carry on; it falls back to working by hand, at a lower rate, for as long as its people and its paper last. During the Collins Aerospace attack of September 2025 (Section 2.3), Brussels, Heathrow and Berlin kept flights moving by manual check-in at a reduced rate, Brussels for about ten days and with handwritten boarding passes. Most of Maersk's terminals fell back to working by hand within two or three days of NotPetya, and Pier 400 had no rate for three (*NotPetya Attack* F-4). Nagoya, with one system for all its terminals, had no fallback at its gate, which stayed shut about three days, while the ministry's committee records ship-side handling by hand from printed work instructions that the harbor transportation association's report does not mention (*Nagoya Ransomware* F-1). The difference was not whether the systems failed but whether the node knew how to work without them, at each of its interfaces, and how fast. The requirements below follow from this finding.

7.3 The Seven Requirements

CDB-1, degradation as baseline. The design basis assumes from the outset that the node's links to its tenants, carriers, agencies and the control class, and the shared platforms that carry many of them, are degraded or lost in any event large enough to matter, and the node's fallback is written and tested against that condition.

CDB-2, local protective and safety autonomy. This requirement is still worded for the electric sector, where it names a grid's supervisory control and energy-management systems. Read for the nodes, it requires that loss of the terminal operating system, the common-use passenger systems, the building management system or the node's networks not cause loss of its power transfer, its life-safety systems or the limits on its equipment. Section 5.6 states the property, and the design basis must demonstrate it rather than assume it.

CDB-3, field-executable emergency procedures. The node's staff must be able to locate and move cargo, process and board passengers, and evacuate the node, from records, forms and procedures held in a form that does not depend on the network. The records held for working by hand are kept current to a stated interval, so that the node can locate what it holds at the hour of

loss and not at the hour of the last copy; after NotPetya some containers were lost track of for months (*NotPetya Attack* F-7).

CDB-4, verified independent channels. The node must be able to reach the control class, its principal tenants and carriers, and the responding agencies over a channel that shares nothing with its primary networks. A node must also be able to learn what it holds — what is in its yard, who is in its terminals — by a means that does not depend on the platform that is lost (Section 7.2). The node must also have a means of reaching the people it holds that shares nothing with its normal power and networks; at Atlanta the public address was lost with the power and staff went through the terminals on foot (*Atlanta Power Outage* F-6). A channel credited under CDB-4 is sized for the traffic of the event in which it is relied on and exercised at that load, because independence from the primary networks is necessary and not sufficient: the national radio network of the Belgian emergency and security services saturated quickly on 22 March 2016 (*Brussels Bombings* F-5). An outside authority's decision to close the node reaches the operator by a direct, acknowledged channel (Section 4.7). And the load lists, bay plans and release information that a diverting node needs reach it independently of the lost platform (Section 3.3).

CDB-5, coordination without shared connectivity. This requirement is also still worded for the electric sector, where it names the gas–electric boundary. Read for the nodes, it requires that the many companies working at a node — terminal operators, airlines, ground handlers, railroads, truckers — be able to work together when they share no network, under a common plan that names who decides what and how they tell each other. The common plan names among its parties the serving utility, with a mechanism to convene the parties agreed in advance, and any company that operates the node's terminals (*Heathrow Closure* F-7; *Atlanta Power Outage* F-6); the public authorities that may order the node's closure (*Brussels Bombings* F-5); and, for a node whose queue waits beyond its fence, the vessel traffic service, the carriers' and terminal operators' associations and the body that keeps the queue (*San Pedro Bay Congestion* F-4). It includes the exchange of load lists, bay plans and release information with the carriers and neighboring nodes that would take the node's flow (*NotPetya Attack* F-5). At Nagoya the association's terminal section served as the body that decided together until the system was restored (*Nagoya Ransomware* F-6).

CDB-6, Command Broker continuity. Sections 6.3 and 6.5 apply it.

CDB-7, verification cadence. Operation with the shared platform lost, evacuation with the node's information systems lost, and the transfer of critical loads on the loss of a supply point are exercised at set intervals, with the primary links assumed lost, and the manual fallback rate is measured in each exercise. The fallback rate is measured for each function and interface, and the exercise trains the staff who would supply it (Section 3.4). The transfer exercised is that for the supply point whose loss takes longest to switch, end to end from the loss to the restart of the last critical system, with the utility taking part and the switchover and restart times measured; Heathrow had never exercised the loss of a single supply point (*Heathrow Closure* F-7). The attack case is among the evacuations exercised (*Brussels Bombings* F-4), and the independent

channels are exercised at the event's load with the discipline of their use (*Brussels Bombings* F-5).

8. Cross-Sector Dependencies and Regulatory Context

8.1 What Other Sectors Depend On

Every sector whose goods move depends on the nodes, because the nodes are where the goods change hands. A port is how fuel, chemicals, food, components and finished goods enter a region; a rail hub is how bulk commodities cross the continent; a cargo airport is how urgent goods and medical supplies move overnight. The framework states the forward-direction elevation for a transportation asset that is the sole movement path for a Tier-2 or Tier-3 facility in another series (framework §6.1). For this class the measure of sole-path status is stranded flow: a node is the sole path for a dependent facility when the share of that facility's supply that would be stranded, if the node were lost, exceeds what the facility can hold in stock for the node's recovery time. The nodes also carry the military-mobility consequence of Section 5.3, which is national rather than economic.

8.2 What the Nodes Depend On

The class depends on electric power above all, because a node without power stops at once and a hub airport's critical systems have no other supply that can carry them; that dependency carries the operate-through requirement of Section 4.1. It depends on the ways for access, on the control class for the authority to move anything beyond its fence, and on the conveyances that call at it. It depends on the communications sector for the networks that join its parties, and on satellite positioning and timing for its automated equipment and its records (Section 4.9). It depends on the water sector for fire protection and for its occupants, on the software suppliers whose platforms it shares, and on the labor of the people who would have to work it by hand. And it depends on the emergency services to answer what happens inside it. For its clearing flow it depends on the drayage, chassis, rail and storage beyond its gate (Section 4.6).

8.3 Boundaries

With Ways and Linehaul. On the author's decision of 8 October 2026, the node stops at its fence. The airspace, the harbor channel and its approaches, the track beyond the yard throat and the road beyond the gate, and the tunnels that join a station to its line are DBA-TRAN-WY-FC2's. On the author's decision of 10 October 2026, the anchorages and holding areas where a node's queue waits are the ways class's as places and the coordination class's to manage, and the queue the node causes there is this class's as a consequence it measures (Section 4.6). The hazard of a queue that fills them, to the conveyances and to submerged infrastructure, is noted to the ways, conveyance and control classes (*San Pedro Bay Congestion* F-3, F-8).

With the coordination class. The control tower, vessel traffic service, rail dispatch and transit control center are DBA-TRAN-CntlCo's wherever they stand, because their consequence is the

authority to move. The node supplies the basis on which that authority is given — whether a berth, gate, runway or platform is ready — and the control class decides. When a hazardous shipment at the node is in trouble, the division of Command Broker authority stated in the framework applies (framework §3.5). Whether the order and place of a queue beyond the fence is an act of movement authority is noted to the control class (*San Pedro Bay Congestion* F-4).

With Conveyances and Fleet. The aircraft at the gate, the vessel at the berth and the train in the yard are DBA-TRAN-CV-FC3's. The node's equipment that serves them — the jet bridge, the crane, the fuel hydrant — is this class's. The people aboard a train at a station's platform are counted within the station's occupancy (Section 3.3). The displacement of conveyances and crews by a node's closure is noted to the conveyance class as a contribution to its fleet unavailability (*Heathrow Closure* F-3).

With the Chemical and Energy series, at port and yard. On the author's decision of 8 October 2026, the release-and-network rule the framework states for tank cars (framework §1.5, §6.3) applies at the node. The harm a release, fire or explosion of hazardous material does is governed by the series that holds it: chemical storage at a port by DBA-CHEM-DI-FC3 (*Distribution, Blending and Supply Terminals*), which states the same rule from its side; loaded chemical containers waiting in a yard by DBA-CHEM-TR-FC2 (*Bulk Chemical Transport*); petroleum terminals and storage by DBA-OIL-FC3 (*Crude and Product Terminals and Storage*); and liquefied natural gas terminals by the gas series. The harm to the node as a transport node is this class's. Both use the shared release model (*Chemical Production and Synthesis* §4). Beirut is read this way: its release belongs to the chemical series, which carries its custody findings in DBA-CHEM-RE-5 (*The West Fertilizer Explosion*), and its loss of the port is this class's reading (Section 4.4).

With Commercial Facilities. On the author's decision of 8 October 2026, where a transit station lies beneath or beside a public-assembly venue, the line is drawn by who acts. The station — its platforms, concourses, stairs, ventilation and the movement of its trains — is this class's, and its plan states what it needs from the venue. The venue above it is the Commercial Facilities series' public-assembly class, which states what the venue needs from the station when it is written. The combined event at the node, which reaches every co-located sector, is the convergence-node analysis the corpus keeps separately (*Commercial Facilities Sector Framework* §1.5).

With Emergency Services. Response to an incident inside the node is the Emergency Services series', and the node supplies the information responders need before they arrive, as the chemical series requires of its terminals (*Distribution, Blending and Supply Terminals* §6.4). An airport's own aircraft rescue and firefighting service is part of the node, and the mutual aid it gives and receives belongs on the mutual-aid map of the Emergency Services series (*Response Forces and Stations* §3.4). In the attack case that information is marked as unverified until it is confirmed, and the attack case, with the responders' way in and out, is in the node's plan (Section 4.7). Both points are noted to *Response Forces and Stations* (*Brussels Bombings* F-4, F-5).

With the electric series. On the author's decisions of 8 and 10 October 2026, the node's power arrangements, critical loads and switchover time are made known to the utility that serves it by a written exchange, the node states what it requires of any supply another party runs inside its fence, and the electric series designs that equipment (Section 4.1). The requirement reaches the utility as quality of service, an agreement or tariff that names the event the node's supply must survive, and not as a raised acceptance criterion of the utility's facilities. The water series' elevation rule, which does raise those criteria for dialysis, fire flow and cooling water, is not applied here, because a node can contract for its supply and those dependents generally cannot (Section 10.3).

With military installations. Military ports of embarkation and air bases are outside the class. A commercial node designated for military use is inside it and carries the presumption of Section 5.1.

8.4 Regulatory and Institutional Context

The nodes are regulated mode by mode. Ports and marine terminals are regulated for security by the Coast Guard under the Maritime Transportation Security Act of 2002 (33 CFR Part 105), and since 16 July 2025 under its rule on cybersecurity in the marine transportation system (33 CFR Part 101, Subpart F), whose cybersecurity plans are due by 16 July 2027. Airports are certificated by the Federal Aviation Administration (14 CFR Part 139) and regulated for security by the Transportation Security Administration (49 CFR Part 1542), which in March 2023 added cybersecurity requirements for airports and aircraft operators by emergency amendment. Freight railroads, passenger railroads and transit are regulated for security by the TSA (49 CFR Parts 1580 and 1582), which has imposed cybersecurity requirements by security directive since December 2021 and proposed to codify them in November 2024, a rule not final as of October 2026; transit and passenger rail stations are designed for fire and egress to NFPA 130. Truck terminals carry no federal facility-security regime of their own; the hazardous-materials security plans of 49 CFR Part 172, Subpart I attach to the shipment and the carrier, not to the terminal. Each regime is mature within its mode. None defines how much flow a node may strand, how fast it must fall back to working by hand, or how long its critical systems may be without power, and none spans the modes. The record of each event confirms the gap. The Coast Guard's rule answers NotPetya's flat network, destroyed systems and supplier in its controls (33 CFR 101.650(f) to (h)), and its requirement of protected and frequently tested backups (33 CFR 101.650(g)) is the regulatory form of the restoration requirement of Section 4.2 for the ports it covers; it sets no fallback rate, restoration time or stranded-flow limit (*NotPetya Attack* F-6, F-9). The Government Accountability Office found that the Federal Aviation Administration does not require backup power for the areas of an airport serving travelers, and the remedy at Atlanta was a settlement, not a requirement (GAO-23-105203; *Atlanta Power Outage* F-9). The Federal Maritime Commission's Fact Finding 29 final report and the National Transportation Safety Board's MIR-24-01 record flow beyond clearing capacity, and no regime sets how long a ship may wait off a node or the capacity of the places where it waits (*San Pedro Bay Congestion* F-9).

Abroad, the review of the Heathrow closure found no explicit cross-sector requirement on critical operators to keep operating through a disruption of power, and the United Kingdom government's response of November 2025 imposes none (*Heathrow Closure* F-9). Japan's measures of 2024 to 2026 regulate the security and supply of a port's terminal operating system, the nearest regime the record shows to Sections 4.2 and 4.8, and set no fallback rate or stranded-flow limit (*Nagoya Ransomware* F-9). That is the gap this class fills (framework §2.3).

9. Inheritance Map and Anticipated Derivatives

9.1 The Inheritance Map

Table 9-1. Inheritance map for DBA-TRAN-ND-FC1.

Element	Source	Inheritance mode
Function-primary taxonomy, mode sub-index, and the class's place in it	framework §3.1, §3.2, §3.6, §8.2	Inherited unchanged
Throughput loss at a concentration point as the governing consequence	framework §3.2	Inherited and developed (Sections 3.2, 3.3)
Consequence-severity inclusion criterion and candidate events	framework §4.1	Inherited and developed (Section 4)
Bright Line and Command Broker architecture	framework §5; FD-BL; FD-BR	Applied; not restated (Section 6)
Determinism as scope	FD-LD §4.3	Applied (Section 6.2)
Advance discharge for repeated automated actions	FD-BL-D1 §4, §7; CB-PAE §4	Applied (Section 6.3)
Indication integrity	FD-BL-D2	Cited (Sections 5.5, 6.3)
Envelope reservation	FD-EV	Applied (Section 1.6)
Liberty consequence of passenger screening	FD-SBL §2.1, §4.4	Cited; the governing Line named per consequence (Section 6.4)
Local sovereignty of power transfer, life-safety systems and equipment limits	FD-LS §3.2	Cited directly (Section 5.6)
Communications design basis CDB-1 to CDB-7	DBA-MA-CDB §2, §3	Cited and applied (Section 7)
Timing receiver design basis	Positioning, Navigation and Timing §7	Cited (Section 4.9)

Element	Source	Inheritance mode
MA applicability gate	DBA-MA-AF §2, §3	Applied, with a presumption for strategic seaports and nodes without alternative (Section 5.1)
Declared residual	DBA-MA-DiD §3.5	Cited directly (Section 5.7)
Release-and-network boundary and the shared release model	framework §1.5, §6.3; Chemical Production and Synthesis §4; Distribution, Blending and Supply Terminals	Applied at the node (Sections 4.4, 8.3)
Sole-path elevation	framework §6.1	Developed; stranded flow as its measure (Section 8.1)
Station and venue boundary	Commercial Facilities Sector Framework §1.5	Stated from this side (Section 8.3)
Independence shown as built, common mode counted as one, and the fallback measured	framework §4.2	Applied (Sections 3.4, 4.1, 4.2, 4.6)
Restoration of sound software to every unit	framework §4.3	Applied to shared platforms (Section 4.2)
Campaign capacity with crew replacement	framework §7.6	Cited for the hold time (Section 3.4)
Stranded flow, clearance time, hold time, notification time, manual fallback rate and hold time, power switchover time and restart time, clearing flow, Sustained Node Condition	—	Defined here

9.2 Anticipated Derivatives

The class anticipates three kinds of derivative. Mode-specific instruments for the hub airport and the major container port, which the framework expects (framework §3.6), would add what is particular to each and cite this document for the rest; their designators are the framework's to confer. The use case for the governing scenario of Section 5.3 is written as DBA-TRAN-UC-1 (*Strategic Seaport Denial*), and Section 10.10 records what it returns. The reference-event instruments for the class's six live events are written and are held by Project 11 (Section 10.5); further events the class proposes are conferred by the framework and written there.

10. Gaps and Open Items

10.1 Quantitative Design Parameters

Stranded flow, clearance time, the manual fallback rate and hold time, and the power switchover time are stated as parameters, on the author's decisions of 8 October 2026. The stranded-flow limit for each kind of node, the switchover time allowed for critical systems, and the fallback rate a node must be able to hold are still to be set, as is how they should differ between a national node and a regional one. The framework adopts the conditionality principle of Section 5.8 for the sector (framework §4.2).

10.2 Design-Basis Durations

The design duration of a campaign against the nodes is not given here. It should be set together with the energy series' assumptions about restoration after a deliberate attack on the grid, the communications series' design outage, and the duration the Emergency Services series sets for its compound event, so that the series assume the same event.

10.3 Power Operate-Through and the Electric Series

Section 4.1 requires that a node's power arrangements, critical loads and switchover time be made known to the utility that serves it, by a written exchange that runs both ways. On the author's decision of 10 October 2026 the question first put to Project 8, whether a node that cannot ride through the loss of a supply point raises the acceptance criteria of the utility facilities that supply it, is withdrawn, and the matter is one of quality of service: the node states its requirement, including what it requires of any supply another party runs inside its fence, and the requirement reaches the utility through an agreement or tariff that names the event (Sections 4.1, 8.3). Who bears the cost of that service is the question registered under Project 9 OI-17, with which OI-27 is joined. The Heathrow review's record that the network companies' licenses generally prohibit discrimination between customers in restoration is noted to Project 8 for information (*Heathrow Closure* F-5).

10.4 Local Sovereignty in the DBA-MA Sense

The protective functions at Section 5.6 cite *Local Sovereignty* §3.2 directly. The endurance requirement beside them is the property the DBA-MA parent framework's local sovereignty principle describes. Whether that principle is FD-LS's construct at all is held pending Project 5's answer, and the requirement is written here in this document's own words until then. The item is registered under Project 9 OI-16.

10.5 Reference Events

Seven events were proposed here and conferred by the framework at its §9.1 as DBA-TRAN-RE-1 to DBA-TRAN-RE-7: the NotPetya attack on Maersk's terminals of June 2017, the ransomware attack on the Port of Nagoya of July 2023, the power outage at Atlanta's airport of

December 2017, the closure of Heathrow after the North Hyde substation fire of March 2025, the bombings at Brussels Airport and Maalbeek station of March 2016, the San Pedro Bay congestion of 2020 to 2022, and the ransomware attack on the Collins Aerospace check-in system of September 2025. On the author's triage ruling of 9 October 2026 the last is withdrawn, its shared-platform findings being held by the NotPetya and Nagoya instruments, and Section 2.3 cites it as an illustration. The six live instruments are written and, under the determination of 9 October 2026, are held by Project 11. This revision takes in what they return: six corrections of the class's account of the events, made at Sections 2.3, 3.2, 3.4, 4.2, 4.6, 5.5, 7.2 and 10.5 and in the References, and the recommendations *NotPetya Attack* R-2 to R-9, *Nagoya Ransomware* R-1 to R-9, *Atlanta Power Outage* R-1 to R-6, R-8 and R-9, *Heathrow Closure* R-1 to R-9, *Brussels Bombings* R-1 to R-8 and *San Pedro Bay Congestion* R-1 to R-9, those of the Atlanta and Heathrow instruments conformed with each other in this one revision as both instruments ask. The accounts of Section 2.3 now rest on the primary records the instruments read: for Nagoya the reports of the ministry's investigating committee and the notices and report of the Nagoya Harbor Transportation Association, beside Lloyd's List and BleepingComputer; for Brussels the summary and reports of the parliamentary inquiry commission and the court's outcome as reported; for Atlanta the releases of Georgia Power, the City and the airport; for Heathrow the review itself; and for San Pedro Bay the Marine Exchange, the Port of Los Angeles, the Federal Maritime Commission and the National Transportation Safety Board. The framework elements of the recommendations are the Transportation Sector Framework's to take in at its next revision, and the elements the instruments note to the conveyance class, the control class and *Response Forces and Stations* are noted there. On the ship working at Nagoya the class records both primary accounts and adopts neither. The Key Bridge collapse belongs to the ways class, which proposed it.

10.6 Two Communications Design Basis Requirements Worded for the Electric Sector

CDB-2 and CDB-5 are read for the nodes at Section 7.3 because their text names the electric sector. When the Military Action Communications Design Basis words them for every sector, Section 7.3 is revised to cite the new wording.

10.7 The Shared Release Model

Sections 4.4 and 8.3 cite the shared chemical-release model at its provisional home, *Chemical Production and Synthesis* §4. When the chemical framework confers a standalone model instrument, this document is among those owed notice that a sweep is due. The item is registered under Project 9 OI-18.

10.8 Screening and the Two Lines

Section 6.4 names, per consequence, which Line governs a screening action, as FD-SBL §2.1 requires. Whether FD-SBL's worked examples or its register should record passenger screening

as a sector application is Project 1's to decide, as it is for the law-enforcement actions of the Emergency Services series. It is put to Project 1 in a note of the same date and registered under Project 9 OI-26.

10.9 Closing and Reopening a Passenger Node

Section 6.3 places the decision to close or reopen a passenger node with the node's operator, because the decision falls on both of the class's chains at once and the operator is the party that holds both. This revision qualifies the placement in three ways from the class's events: where a public authority may close the node by law, both act and neither waits (*Brussels Bombings* F-3); the holders of the decisions to close, to hold or clear, and to reopen are named in advance with their succession, and reopening may be by part (*Heathrow Closure* F-6; *Atlanta Power Outage* F-5); and the resumption of work on a restored platform is a reopening (*Nagoya Ransomware* F-6). The control class, now written, supplies the basis for movement and states the rule for a public authority at its §6.3. Whether the control class or the responding agencies should hold any part of the decision itself, rather than supply its basis, remains for the author, and Brussels is the case for that decision. The framework's placement of a public authority's closure in the control class (framework §5.5) is noted to its next revision, since the authorities that closed the Brussels metro stood outside it.

10.10 What the Use Case Returns

DBA-TRAN-UC-1 (*Strategic Seaport Denial*) runs the governing scenario of Section 5.3 through this class. It validates the class in two respects, the fallback rate stated for the military-mobility flow with platform, links and a supply point lost together, and the residual set by the campaign rather than the port, and returns three gaps, taken in at this revision: the time to reconcile suspect records with the physical stock and the rate for flow whose records are suspect (Sections 3.4, 5.6); the hold time reckoned with the replacement of the crews who would work by hand (Section 3.4); and the decision between restoring a platform and preserving it, and the check before work resumes on it (Section 6.3). Each joins a reference event's return, Nagoya's F-7, F-2 and F-6 respectively. The hold time cannot be judged until the campaign's duration is set, which stays open under Section 10.2.

References

DBA-TRAN Series and Parent Framework

Transportation Sector Framework (DBA-TRAN_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

Ways and Linehaul (DBA-TRAN-WY-FC2), *Conveyances and Fleet* (DBA-TRAN-CV-FC3) and *Traffic Management, Signaling and Control* (DBA-TRAN-CntlCo). Eclectic Technologies, 2026. The sibling classes; the control class's authorization map and closure rule (§3.3, §6.2, §6.3).

The NotPetya Attack on Maersk's Terminals, June 2017 (DBA-TRAN-RE-1), *The Ransomware Attack on the Port of Nagoya, July 2023* (DBA-TRAN-RE-2), *The Power Outage at Hartsfield-Jackson Atlanta International Airport, December 2017* (DBA-TRAN-RE-3), *The Closure of Heathrow Airport, March 2025* (DBA-TRAN-RE-4), *The Bombings at Brussels Airport and Maalbeek Station, March 2016* (DBA-TRAN-RE-5) and *The Congestion of the San Pedro Bay Ports, 2020 to 2022* (DBA-TRAN-RE-6). Eclectic Technologies, 2026, held by Project 11. The class's reference-event instruments, cited by the short titles *NotPetya Attack*, *Nagoya Ransomware*, *Atlanta Power Outage*, *Heathrow Closure*, *Brussels Bombings* and *San Pedro Bay Congestion* and finding or recommendation number.

Strategic Seaport Denial (DBA-TRAN-UC-1). Eclectic Technologies, 2026. The use case for the governing scenario (Section 10.10).

DBA-MA Parent Framework. Eclectic Technologies. Source of the four military-action characteristics.

Military Action Communications Design Basis (DBA-MA-CDB). Eclectic Technologies, 2026. CDB-1 through CDB-7 (§2) and the instantiation requirement (§3).

MA Applicability Framework (DBA-MA-AF). Eclectic Technologies, 2026. The four-filter applicability gate (§2, §3).

Defense in Depth (DBA-MA-DiD). Eclectic Technologies, 2026. Declared-residual form (§3.5).

Other Series

Chemical Production and Synthesis (DBA-CHEM-PR-FC1). Eclectic Technologies, 2026. The shared chemical-release model, held provisionally at §4.

The West Fertilizer Explosion, April 2013 (DBA-CHEM-RE-5). Eclectic Technologies, 2026, held by Project 11. Material awaiting disposal and the custody finding (F-11, F-12).

Bulk Chemical Transport (DBA-CHEM-TR-FC2). Eclectic Technologies, 2026. Loaded containers waiting in yards.

Distribution, Blending and Supply Terminals (DBA-CHEM-DI-FC3). Eclectic Technologies, 2026. Chemical storage at ports; hazard information for responders (§6.4).

Crude and Product Terminals and Storage (DBA-OIL-FC3). Eclectic Technologies, 2026. Petroleum terminals and storage.

Positioning, Navigation and Timing (DBA-COMM-PN-FC4). Eclectic Technologies, 2026. The timing receiver design basis (§7).

Response Forces and Stations (DBA-ESS-RF-FC2). Eclectic Technologies, 2026. Two Lines for law-enforcement actions (§6.4).

Commercial Facilities Sector Framework (DBA-CF_Sector_Framework). Eclectic Technologies, 2026. The convergence node (§1.5).

Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*) and CB-PAE (*Pre-Approved Action Envelope*), cited by designation; version state in CB-REG (*Command Broker Family Register*). FD-SBL (*The Surveillance Bright Line*), cited by designation; version state in FD-SBL-REG (*Surveillance Bright Line Register*).

Regulatory and Standards

Maritime Transportation Security Act of 2002; 33 CFR Part 105, Maritime Security: Facilities; U.S. Coast Guard, *Cybersecurity in the Marine Transportation System*, final rule, 90 FR 6298, 17 January 2025, codified at 33 CFR Part 101, Subpart F, including 33 CFR 101.650(f) to (h). 14 CFR Part 139, Certification of Airports; 49 CFR Part 1542, Airport Security; Transportation Security Administration, cybersecurity emergency amendment for airport and aircraft operators, 7 March 2023. 49 CFR Part 1580, Freight Rail Transportation Security; 49 CFR Part 1582, Public Transportation and Passenger Railroad Security; TSA Security Directives 1580-21-01 and 1582-21-01, December 2021; TSA, *Enhancing Surface Cyber Risk Management*, proposed rule, 89 FR 88488, 7 November 2024. 49 CFR Part 172, Subpart I, Safety and Security Plans. NFPA 130, *Standard for Fixed Guideway Transit and Passenger Rail Systems*. Maritime Administration, *Commercial Strategic Seaports* (National Port Readiness Network).

Government Accountability Office, GAO-23-105203, on backup power at airports. Federal Maritime Commission, Fact Finding 29, final report. National Transportation Safety Board, MIR-24-01. Government of the United Kingdom, response to the review of the North Hyde substation outage, November 2025. Japan’s measures of 2024 to 2026 on the security and supply of port terminal operating systems, as *Nagoya Ransomware* records them.

Data and Events

Bureau of Transportation Statistics, *Port Performance Freight Statistics Program*, 2025 and 2026 Annual Reports. Federal Aviation Administration, CY2024 commercial service enplanements (September 2025), and *Terminal Area Forecast Summary FY2025–FY2055*. Federal Highway Administration, *Talking Freight* webinar, 19 October 2016; CREATE Program, *Overview*, April 2021; *Trains*, the decline of North America’s hump yards, December 2024. Airports Council International, preliminary world airport rankings for 2024.

A.P. Moller-Maersk, *Interim Report Q2 2017*, 16 August 2017, and its announcement of 28 June 2017; APM Terminals, update of 30 June 2017; US-CERT, TA17-181A; Talos; the governments’ attribution statements of 15 and 16 February 2018; Port Strategy, Seatrade Maritime and FOX 11 Los Angeles reporting of the APM Terminals closures, June and July 2017.

Ministry of Land, Infrastructure, Transport and Tourism, reports of the committee investigating the Nagoya United Terminal System outage, 2023; Nagoya Harbor Transportation Association,

notices of July 2023 and report of 26 July 2023; official trade statistics for 2023; Lloyd's List and BleepingComputer reporting of the attack, July 2023.

Georgia Power, releases of 17 and 18 December 2017; City of Atlanta, release of 19 December 2017; Hartsfield-Jackson Atlanta International Airport, statement of 20 December 2017; CBS News and the Associated Press, and WSB-TV, reporting of the outage, 18 December 2017.

National Energy System Operator, *Final report from the review into the North Hyde Substation outage*, dated 30 June 2025, published 1 July 2025, and its news page of 1 July 2025; evidence of the airline operators' representative and of the airport's chief executive to the parliamentary committee; Press Association reporting of the disruption.

Belgian Chamber of Representatives, parliamentary inquiry commission on the attacks of 22 March 2016, summary and reports DOC 54 1752/006, /007 and /010; the court's outcome of 2023, as reported; PBS NewsHour, 1 May 2016, and Agence France-Presse, 16 April 2016, reporting of the airport's reopening; VRT reporting of the airport's 2016 traffic.

Marine Exchange of Southern California, release of 11 November 2021; Port of Los Angeles, container figures and release of 25 October 2021; Fresh Fruit Portal, July 2021, reporting the Marine Exchange's statement to Insider; Associated Press reporting; Federal Maritime Commission, Fact Finding 29 final report; National Transportation Safety Board, MIR-24-01; Maritime Executive, 22 November 2022, quoting the Marine Exchange on the end of the backlog.

Travel Extra, VRT and BX1 reporting of the Collins Aerospace MUSE attack, September 2025.

U.S. Army Corps of Engineers, restoration of the Fort McHenry Federal Channel, 10 June 2024; Office of the Governor of Maryland, Port of Baltimore cargo records, 23 February 2024.

United Nations officials, including the World Food Programme, as reported by Reuters, September 2020, on the Port of Beirut's grain and bulk capacity.

Revision History

Version	Date	Author	Description
0.1 — DRAFT	October 2026	Tim Roxey	Initial issue. The first facility class of the DBA-TRAN series, written first on the author's decision of 8 October 2026 that the classes follow the register's order. One class sub-indexed by mode, typed physical, with the node bounded at its fence, on the author's decisions of the same date. Defines two consequence chains, throughput and occupancy, carried as each node's function requires, with stranded flow and clearance time as the governing measures and the manual fallback rate and hold time and the power switchover time as further parameters. Makes the loss of a shared software platform a design-basis event of its own and requires an operate-through power arrangement known to the serving utility.

Version	Date	Author	Description
			Applies the release-and-network rule to petroleum, gas and chemical terminals and to hazardous cargo at the node, and draws the line with Commercial Facilities at a station beneath a venue by who acts. Presumes the military-action determination positive for designated strategic seaports and for nodes without an absorbing alternative, with the denial of a strategic seaport during a deployment as the governing scenario. Names FD-BL for the security consequence and FD-SBL for the liberty consequence of passenger screening. Proposes seven reference events for the framework to confer and cites DBA-CHEM-RE-6. Registers the power question for the electric series under Project 9 OI-27, the endurance requirement under OI-16, the release model under OI-18 and the screening question under OI-26. An independent check of the text against its sources raised 13 items, 13 taken before issue.
0.2 — DRAFT	October 2026	Tim Roxey	Takes in what the class's six live reference-event instruments return, now held by Project 11: six corrections of the class's account of the events, at Sections 2.3, 3.2, 3.4, 4.2, 4.6, 5.5, 7.2 and 10.5 and in the References, and *NotPetya Attack* R-2 to R-9, *Nagoya Ransomware* R-1 to R-9, *Atlanta Power Outage* R-1 to R-6, R-8 and R-9, *Heathrow Closure* R-1 to R-9, *Brussels Bombings* R-1 to R-8 and *San Pedro Bay Congestion* R-1 to R-9, the Atlanta and Heathrow recommendations conformed in one revision. Stranded flow counts flow delayed and is counted at each interface; recovery time runs to the return of flow; diversion is credited only with its own information (§3.3). The fallback rate is stated by interface and function, with the people who supply it, and the switchover time is defined and measured, with a time to restore by repair where no transfer remains; a parameter for the measures on a queue is added (§3.4). Redundancy of supply is shown by independence along the whole route and by the ability to carry another's load, point-of-use supplies carry public address and vertical circulation for the longer of clearance and switchover, and the knowledge exchange with the utility runs both ways (§4.1). The operator's network and identity domain are shared platforms, platform ownership is contracted, and restoration and backups are required (§4.2); the clearing flow, the queue and diversion under surge are stated (§4.6); the attack on the crowd moves into §4.7 with a notification time and the attack case in the plan; the node's systems are separated from the enterprise network and remote access is required (§4.8). The military-action extension takes the collateral reach, the act disguised as failure, the adversary's supply point, the sequence across nodes, alteration and reconciliation (§5.2 to §5.6), and declares a third residual, for the people a passenger node holds (§5.7). Section 6.3 names the holders of closing, holding, reopening by part, isolation and restoration, with

Version	Date	Author	Description
			the public authority's power to close. Section 7.3 reads CDB-3, CDB-4, CDB-5 and CDB-7 accordingly. On the author's decisions of 10 October 2026: the queue is the node's and the anchorage the ways class's; train occupants at a platform count in the station's occupancy; a hub airport whose flow cannot be diverted falls under the presumption of §5.1; the node states what it requires of a supply another party runs inside its fence, which the electric series designs, and the requirement reaches the utility as quality of service (§4.1, §8.3, §10.3; OI-27 joined to OI-17); Japan's measures are cited at §8.4; and the residual for people held is declared. The Collins Aerospace event is withdrawn as a reference event and cited as an illustration; the citation of Beirut moves to DBA-CHEM-RE-5 (*The West Fertilizer Explosion*). Section 10.10 records what DBA-TRAN-UC-1 returns, and §4.5 records that the ways class proposed the Key Bridge collapse. An independent check of the revision against the six instruments, their errata and the use case raised 10 items: 9 taken before issue, and one, the citation of West F-12 and of Beirut's withdrawal, confirmed against Project 11's batch 15 note and Project 0's notice of the triage rulings. Issued under framework v0.10. Revision-history entries are left as records.
0.3 — DRAFT	October 2026	Tim Roxey	Statements about the framework's state conformed to framework v0.11, which adopts the conditionality principle at §4.2: §5.8 and §10.1 record the adoption in place of the request, and §1.6 no longer describes framework §4, §5 and §7 as skeletons. No other text is changed; the text of v0.2 and v0.3 was compared and differs only at these sites. Issued under framework v0.11. Revision-history entries are left as records.